

Quantum control for quantum information transfer mediated by topological edge states

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We study quantum state transfer in one-dimensional topological chains based on the Su-Schrieffer-Heeger (SSH) model, comparing two distinct paradigms: domain wall-mediated transfer and counterdiabatic (CD) driving. First, we numerically reproduce a domain wall transfer protocol, achieving fidelities $f > 0.999$ for symmetric two-domain configurations with $L = 11$ sites. We investigate the effect of asymmetric domain wall placement, demonstrating that the fidelity is constrained by the exponential sensitivity of the effective couplings to the domain lengths—a limitation of the static Rabi transfer protocol of Ref. [1] rather than of topological protection itself. We further explore shortcut-to-adiabaticity (STA)-inspired ramp shaping—pulse profiles designed to suppress diabatic excitations and emulate adiabatic evolution in finite time—and find that no pulse optimization can overcome the coupling asymmetry that governs the transfer phase. Second, we present an extensive analysis of a hybrid analog-digital protocol for nonadiabatic edge-state transfer, where the SSH Hamiltonian is supplemented with approximate counterdiabatic interactions derived from adiabatic gauge potentials and optimized via variational quantum circuits. Third, motivated by the CTAP (Coherent Tunnelling Adiabatic Passage) formalism, we replace the static Rabi protocol by a dynamic counter-intuitive pulse sequence that exploits the topological dark state of the effective three-level Hamiltonian. Numerical integration of the full $L = 21$ chain recovers fidelities $f > 0.999$ at wall positions where the Rabi protocol collapses, with bulk leakage below 10^{-7} and an effective three-level reduction that is quantitatively exact; the residual infidelity is a removable pulse boundary-condition effect, not a topological bound. A collective version of the protocol transfers across a multi-domain chain with fidelity 0.998 while shortening the transfer time by more than four orders of magnitude. Crucially, because CTAP uses only nearest-neighbor couplings it inherits the chiral protection that pins the edge states against off-diagonal disorder, whereas the next-nearest-neighbor driving of the CD protocol breaks that symmetry: the control protocol, not the topology alone, determines whether the transfer is robust. We also discuss the even-state parity constraint limiting the two-domain-wall case.

I. INTRODUCTION

Quantum state transfer—the coherent transport of quantum information between distant nodes of a quantum network—is a fundamental primitive for quantum computing and communication architectures [2–4]. In solid-state platforms such as quantum dot arrays, superconducting circuits, or cold-atom lattices, the transfer must contend with disorder, decoherence, and the difficulty of precisely engineering long-range couplings.

A promising approach exploits the *topological protection* of edge states in one-dimensional lattice models [5, 6]. The Su-Schrieffer-Heeger (SSH) model [7, 8]—a paradigmatic example of a symmetry-protected topological insulator in class BDI [9]—supports exponentially localized zero-energy edge states when the system is in the topological phase. These states are robust against local perturbations that respect the protecting chiral symmetry, making them attractive candidates as quantum information carriers.

J. Zurita, C. E. Creffield, and G. Platero [1] demonstrated that chains comprising multiple SSH domains separated by *topological domain walls* can mediate fast, high-fidelity quantum transfer. The key mechanism is an exponential speed-up arising from the multi-domain structure: by distributing N domain walls along the chain, the transfer time scales as $(w/v)^{\ell/2}$ per domain

(where ℓ is the domain length and v/w the hopping ratio) rather than $(w/v)^{L/2}$ for the entire chain. This exponential compression of the transfer time, combined with the topological robustness of the boundary states, provides a powerful paradigm for quantum information transport.

In this work, we present a study on topological quantum transfer in SSH chains. We focus on two-domain ($N = 2$) SSH chains and address three questions:

1. Can the transfer protocol of Ref. [1] be faithfully reproduced numerically, and what are the optimal parameters?
2. How does the transfer fidelity degrade when the domain wall is displaced from the chain center, breaking the symmetry between the two domains?
3. Can shortcut-to-adiabaticity (STA)-inspired ramp shaping (pulse-profile optimization of the preparation/readout ramps) [10–12] improve the transfer performance, particularly in asymmetric configurations?

Additionally, we provide a comprehensive analysis of an alternative approach: the hybrid analog-digital protocol recently proposed by S. V. Romero, X. Chen, G. Platero, and Y. Ban [13], where nonadiabatic transfer is achieved through counterdiabatic (CD) driving rather than topological domain walls. We compare both

paradigms in terms of their physical mechanisms, scalability, and robustness.

Finally, motivated by the CTAP framework of A. D. Greentree, J. H. Cole, A. R. Hamilton, and L. C. L. Hollenberg [14], we show how the limitations of the static Rabi protocol in asymmetric configurations can be overcome by means of a dynamic counter-intuitive pulse sequence that exploits the topological dark state of the effective three-level Hamiltonian.

The remainder of this document is organized as follows. Section II introduces the SSH model, its topological properties, and the domain wall construction. Section III describes the transfer protocol and presents our reproduction of the main results of Ref. [1]. Section IV analyzes the effect of asymmetric domain wall placement. Section V investigates STA-inspired ramp shaping. Section VI presents a detailed exposition of the hybrid analog-digital counterdiabatic protocol. Section VII provides a systematic comparison of the domain wall and counterdiabatic approaches. Section VIII introduces the CTAP framework as a route beyond the static Rabi limitation, analyzes the topological dark state, and discusses extensions to two-domain wall systems. Section IX summarizes our findings and outlines future directions.

II. THE SSH MODEL AND DOMAIN WALLS

A. The Su-Schrieffer-Heeger Hamiltonian

The SSH model describes a one-dimensional tight-binding chain with alternating hopping amplitudes v (intracell) and w (intercell) [7, 8]. For a chain of L sites labeled $j = 0, 1, \dots, L-1$, the Hamiltonian reads

$$\mathcal{H} = - \sum_{j=0}^{L-2} t_j (|j\rangle \langle j+1| + |j+1\rangle \langle j|), \quad (1)$$

where

$$t_j = \begin{cases} v & \text{if } j \text{ is even (intracell),} \\ w & \text{if } j \text{ is odd (intercell).} \end{cases} \quad (2)$$

The unit cell contains two sites—sublattice a (even sites) and sublattice b (odd sites). In the thermodynamic limit ($L \rightarrow \infty$), the Bloch Hamiltonian in momentum space is $\mathcal{H}(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}$, where $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$ and the winding of $\mathbf{d}(k)$ around the origin classifies the topological phases.

B. Energy spectrum and topological transition

The bulk dispersion relation is

$$E(k) = \pm \sqrt{v^2 + w^2 + 2vw \cos k}, \quad (3)$$

yielding a spectral gap

$$\Delta = 2|w - v| \quad (4)$$

that closes at $v = w$, signaling the topological phase transition. For $v < w$ the system is in the *topological phase* with winding number $\nu = 1$, while for $v > w$ it is in the *trivial phase* with $\nu = 0$ [6, 15].

Figure 1 shows the energy spectrum of a finite SSH chain ($L = 10$) as a function of v/w . The gap closes linearly at the critical point $v = w$. For a chain with an odd number of sites ($L = 11$), a zero-energy edge state appears in the topological phase, manifesting the bulk-boundary correspondence [16].

C. Topological invariant

The topological classification of the SSH model is captured by the *winding number*

$$\nu = \frac{1}{2\pi} \oint dk \frac{d}{dk} \arg[d_x(k) + i d_y(k)], \quad (5)$$

where $d_x(k) = v + w \cos k$ and $d_y(k) = w \sin k$. The winding number counts how many times the vector $\mathbf{d}(k)$ encircles the origin as k traverses the Brillouin zone. For the SSH model:

- $v < w$: $\mathbf{d}(k)$ encircles the origin $\Rightarrow \nu = 1$ (topological),
- $v > w$: $\mathbf{d}(k)$ does not encircle the origin $\Rightarrow \nu = 0$ (trivial).

The winding number is quantized and can only change when the gap closes ($v = w$), reflecting its topological nature. By the bulk-boundary correspondence, a chain with $\nu = 1$ hosts one protected zero-energy state per boundary.

D. Domain walls and protected states

A *domain wall* (DW) is created at the interface between two SSH domains with different dimerization patterns. Consider a chain of total length L composed of two domains meeting at site j_{DW} :

- **Domain 1** ($j < j_{\text{DW}}$): standard SSH hopping pattern $(v, w, v, w, \dots)$,
- **Domain 2** ($j \geq j_{\text{DW}}$): the pattern restarts as $(v, w, v, w, \dots)$.

When j_{DW} is at an *odd* site, the bond at the interface involves two consecutive v -bonds (a vv defect), which constitutes a topological domain wall. This is because the dimerization pattern on each side of the interface is conjugate: if the left domain is in the topological phase, the right domain effectively presents the reversed pattern.

For a chain with $N = 2$ domains (one domain wall), the system hosts three topologically protected states near

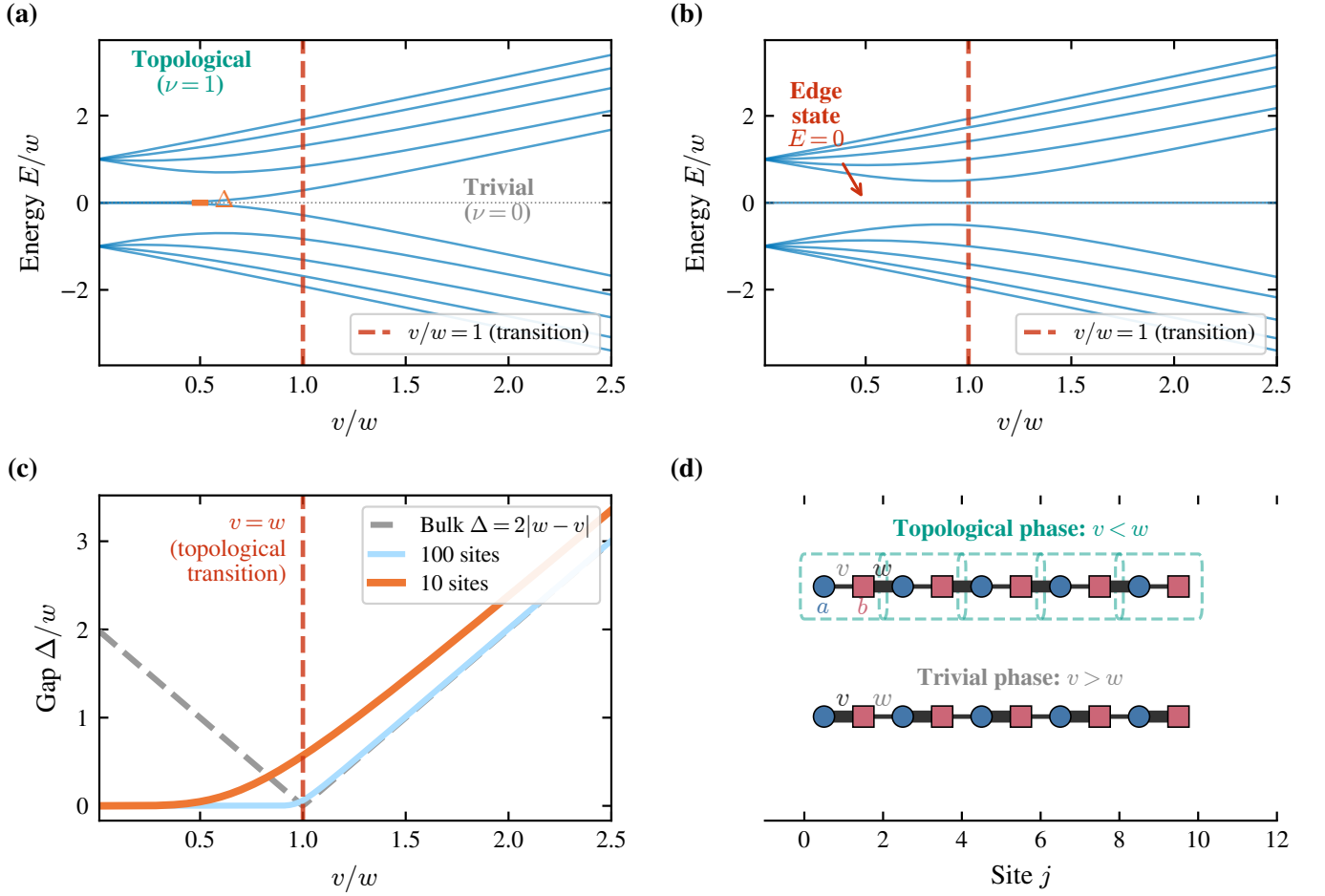


FIG. 1. Energy spectrum of the SSH dimer chain as a function of the hopping ratio v/w . **Left:** Even chain ($L = 10$). The bulk gap $\Delta = 2|w - v|$ closes at $v = w$. **Right:** Odd chain ($L = 11$). A zero-energy state (red) appears in the topological phase ($v/w < 1$), localized at the boundary.

$E = 0$ (Fig. 2):

$$|\mathcal{L}\rangle \propto \sum_{n=0}^{\ell_1/2} (-r)^n |2n\rangle, \quad (6)$$

$$|\mathcal{S}\rangle \propto |j_{\text{DW}}\rangle + \sum_{n=1}^{\ell_1/2} (-r)^n |j_{\text{DW}} - 2n\rangle + \sum_{n=1}^{\ell_2/2} (-r)^n |j_{\text{DW}} + 2n\rangle, \quad (7)$$

$$|\mathcal{R}\rangle \propto \sum_{n=0}^{\ell_2/2} (-r)^n |L - 1 - 2n\rangle, \quad (8)$$

where $r = v/w$ is the decay ratio, and ℓ_1, ℓ_2 are the lengths (number of bonds) of the left and right domains, respectively. Each state resides on a single sublattice and decays exponentially away from its localization center with characteristic length $\xi = -1/\ln r$.

E. Effective low-energy Hamiltonian

The three protected states span an effective low-energy subspace described by

$$\mathcal{H}_{\text{eff}} = J_{\mathcal{LS}} |\mathcal{S}\rangle\langle\mathcal{L}| + J_{\mathcal{SR}} |\mathcal{R}\rangle\langle\mathcal{S}| + \text{H.c.}, \quad (9)$$

which has the structure of a three-site tight-binding chain. The effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ depend on the overlap of adjacent protected states through the SSH bulk. From the analytical expressions [Eqs. (6)–(8)], these couplings scale as [1]

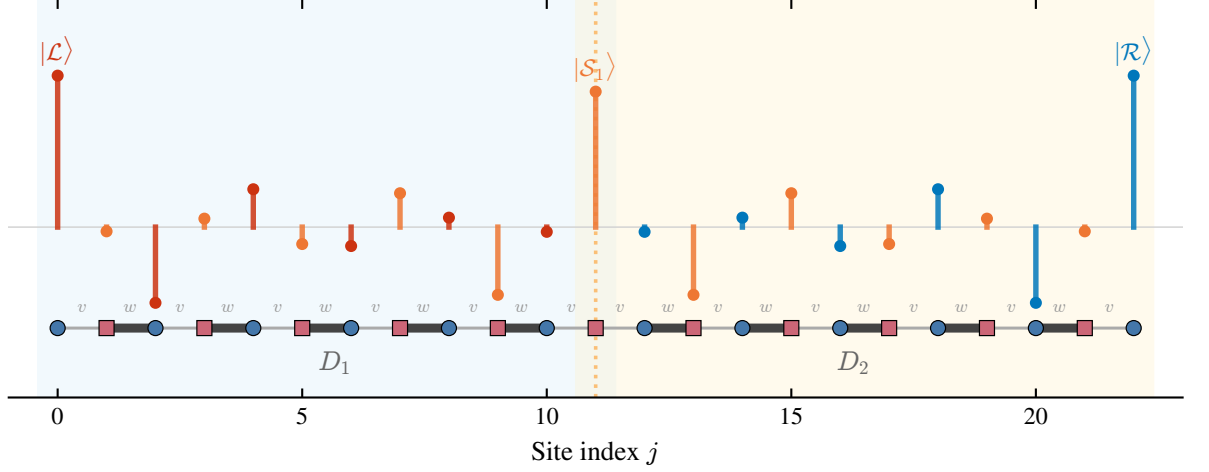
$$J \sim v \cdot \mathcal{M}_i \cdot \mathcal{M}_j \cdot \left(\frac{v}{w}\right)^{\ell/2}, \quad (10)$$

where ℓ is the domain length separating the two states, and the normalization constants are

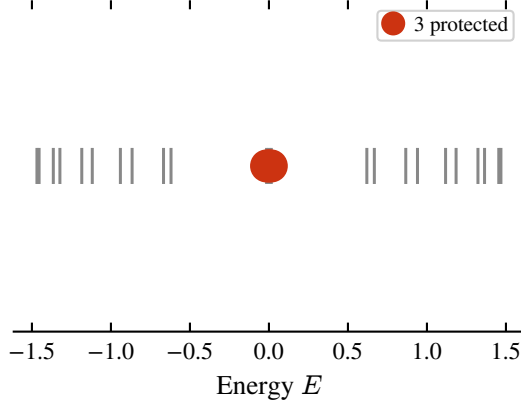
$$\mathcal{M}_{\mathcal{L}} = \mathcal{M}_{\mathcal{R}} = \sqrt{\frac{w^2}{v^2} - 1}, \quad \mathcal{M}_{\mathcal{S}} = \sqrt{\frac{w^2 - v^2}{w^2 + v^2}}. \quad (11)$$

Crucially, the exponential dependence of J on ℓ implies that even a small asymmetry $\ell_1 \neq \ell_2$ leads to vastly

(a)



(b)



(c)

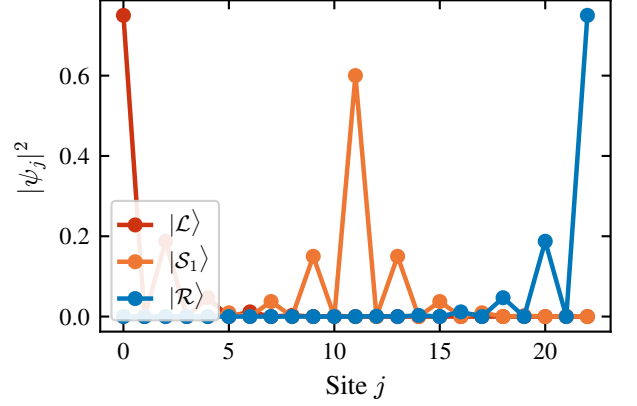


FIG. 2. Topologically protected states in a two-domain SSH chain ($L = 23$, $N = 2$, $\ell = 10$, $v = 0.5$, $w = 1.0$). The three states $|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, and $|\mathcal{R}\rangle$ are localized at the left edge, the domain wall, and the right edge, respectively. Sublattice polarization (alternating signs) is clearly visible.

different effective couplings:

$$\frac{J_{\mathcal{L}\mathcal{S}}}{J_{\mathcal{S}\mathcal{R}}} \sim \left(\frac{w}{v}\right)^{(\ell_2 - \ell_1)/2}. \quad (12)$$

For our parameters $w = 2v$, a difference $|\ell_1 - \ell_2| = 2$ already produces a coupling ratio of 4, while $|\ell_1 - \ell_2| = 8$ yields a ratio of $2^4 = 16$.

III. TRANSFER PROTOCOL

A. Adiabatic preparation and readout

The transfer protocol proposed by J. Zurita, C. E. Creffield, and G. Platero [1] consists of three phases:

1. **Preparation** ($0 \leq t < t_{\text{prep}}$): The intracell hopping is ramped from $v = 0$ (fully dimerized) to $v = v_{\text{tr}}$ following a smooth pulse shape. This adi-

abatically maps the initial localized state $|0\rangle$ onto the protected boundary state $|\mathcal{L}\rangle$.

2. **Transfer** ($t_{\text{prep}} \leq t < t_{\text{tr}} - t_{\text{prep}}$): The hopping is held constant at $v = v_{\text{tr}}$. The effective couplings $J_{\mathcal{L}\mathcal{S}}$, $J_{\mathcal{S}\mathcal{R}}$ drive Rabi-like oscillations in the protected subspace, transferring population from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$ via $|\mathcal{S}\rangle$.

3. **Readout** ($t_{\text{tr}} - t_{\text{prep}} \leq t \leq t_{\text{tr}}$): The hopping is ramped back to $v = 0$, adiabatically mapping $|\mathcal{R}\rangle$ back onto the localized state $|L-1\rangle$.

We refer to this scheme as the *static Rabi protocol*: “static” because the effective couplings $J_{\mathcal{L}\mathcal{S}}$, $J_{\mathcal{S}\mathcal{R}}$ are held constant during the transfer phase, and “Rabi” because the population then undergoes a Rabi-like oscillation $|\mathcal{L}\rangle \rightarrow |\mathcal{S}\rangle \rightarrow |\mathcal{R}\rangle$. This naming distinguishes it from the time-modulated (“dynamic”) pulse sequences considered in Sec. VIII.

The standard pulse shape (Eq. (11) of Ref. [1]) is

$$v(t) = \begin{cases} v_{\text{tr}} \sin^2(\Omega t) & 0 \leq t < t_{\text{prep}}, \\ v_{\text{tr}} & t_{\text{prep}} \leq t < t_{\text{tr}} - t_{\text{prep}}, \\ v_{\text{tr}} \sin^2(\Omega(t - t_{\text{tr}})) & t_{\text{tr}} - t_{\text{prep}} \leq t \leq t_{\text{tr}}, \end{cases} \quad (13)$$

with $\Omega = \pi/(2t_{\text{prep}})$. The $\sin^2$ ramp is chosen for its smoothness: both $v(t)$ and $\dot{v}(t)$ vanish at $t = 0$ and $t = t_{\text{tr}}$, minimizing diabatic transitions to bulk states. The adiabatic condition requires $t_{\text{prep}} \gg \tau = 2/\Delta$, where Δ [Eq. (4)] is the instantaneous spectral gap.

The transfer fidelity is defined as the occupation probability of the target site at the end of the protocol:

$$f = |\langle L-1 | \psi(t_{\text{tr}}) \rangle|^2. \quad (14)$$

B. Transfer time

For a symmetric two-domain chain ($\ell_1 = \ell_2 = \ell$), the effective Hamiltonian (9) has equal couplings $J_{\mathcal{LS}} = J_{\mathcal{SR}} \equiv J$. The time for complete population transfer from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$ via the intermediate state $|\mathcal{S}\rangle$ in a uniform three-site chain is

$$t_{\text{tr}}^{\text{eff}} = \frac{\pi}{\sqrt{2}J}, \quad (15)$$

which, using Eq. (10), scales exponentially with the domain length:

$$t_{\text{tr}} \sim \frac{\pi(w^2 + v^2)}{2(w^2 - v^2)} \left(\frac{w}{v}\right)^{\ell/2+1}. \quad (16)$$

C. Numerical results: reproducing Ref. [1]

We simulate the full time-dependent Schrödinger equation for a chain of $L = 11$ sites ($N = 2$ domains, $\ell = 4$) with $w = 1.0$, $v_{\text{tr}} = 0.5$, and $t_{\text{prep}} = 15$. The time evolution is performed via exact exponentiation of the Hamiltonian at each time step $\Delta t = 0.5$.

Figure 3 shows the transfer dynamics for $t_{\text{tr}} = 45.0$. The initial state $|0\rangle$ is clearly seen to evolve through the three protected states: population flows from the left edge to the domain wall and then to the right edge, achieving a final fidelity $f = 0.999$.

Figure 4 presents the fidelity as a function of t_{tr} , showing Rabi-like oscillations with the first maximum at $t_{\text{tr}} \approx 45$, in excellent agreement with the theoretical prediction $t_{\text{tr}} \approx 45.6$ from Eq. (16) and the value reported in Ref. [1]. The peak fidelity $f = 0.999$ exceeds $f_0 = 0.995$, which we adopt throughout as a stringent high-fidelity benchmark, motivated by values often discussed in the context of fault-tolerant quantum error correction [17].

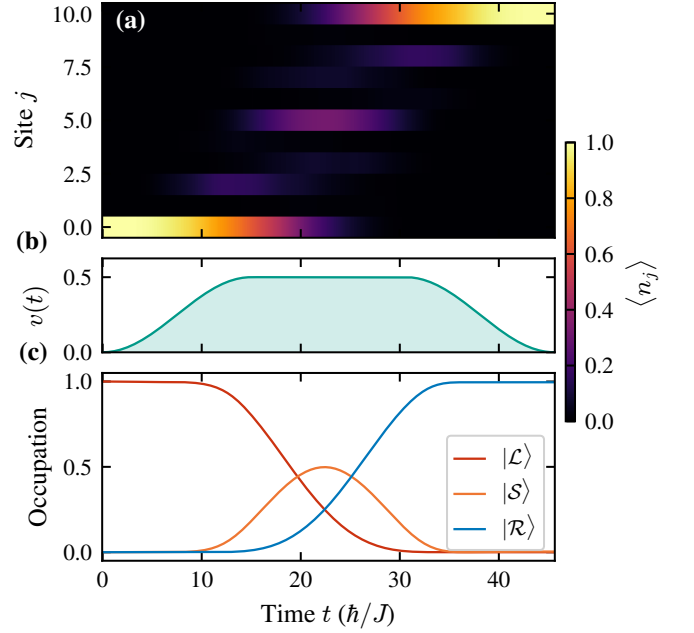


FIG. 3. Transfer protocol dynamics for the symmetric two-domain SSH chain ($L = 11$, $\ell = 4$, $v_{\text{tr}} = 0.5$, $w = 1.0$, $t_{\text{prep}} = 15$, $t_{\text{tr}} = 45$). Population is transferred from site 0 to site 10 via the topologically protected subspace, with final fidelity $f = 0.999$.

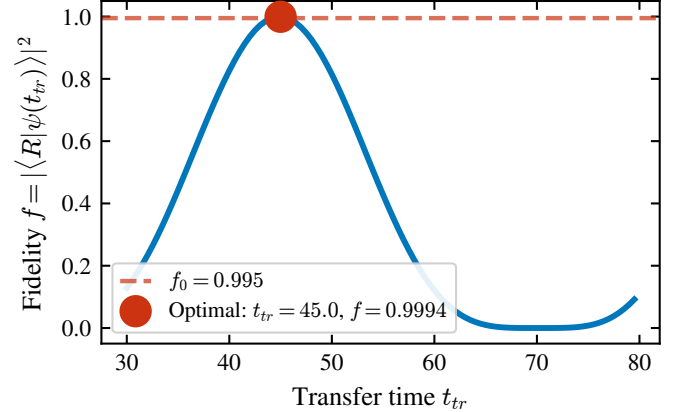


FIG. 4. Transfer fidelity as a function of the total protocol time t_{tr} for the symmetric chain ($L = 11$, $\ell = 4$). The first maximum occurs at $t_{\text{tr}} \approx 45$ with $f = 0.999$, consistent with Ref. [1]. Subsequent oscillations arise from the Rabi dynamics of the effective three-site chain.

IV. ASYMMETRIC DOMAIN WALL PLACEMENT

A. Motivation

In practice, fabrication imperfections may shift the domain wall away from the ideal center of the chain. Furthermore, for chains of certain lengths, it is geometrically impossible to place the domain wall exactly at the center.

The domain wall position j_{DW} must be at an *odd-index* site for the vv -bond defect to occur; thus, only chains with $L = 4k + 3$ (i.e., $L = 11, 15, 19, 23, \dots$) admit a perfectly centered domain wall with $\ell_1 = \ell_2$.

We study a chain of $L = 21$ sites (10 complete unit cells plus one boundary site), varying the domain wall position among three representative odd sites:

- $j_{\text{DW}} = 11$ (quasi-center): $\ell_1 = 11, \ell_2 = 9$,
- $j_{\text{DW}} = 7$ (one-third): $\ell_1 = 7, \ell_2 = 13$,
- $j_{\text{DW}} = 5$ (one-quarter): $\ell_1 = 5, \ell_2 = 15$.

B. Protected states in asymmetric configurations

Regardless of the domain wall position, the system always hosts three protected states near $E = 0$, given by Eqs. (6)–(8) with $\ell_1 \neq \ell_2$. We verify this numerically by diagonalizing the full Hamiltonian and comparing the three eigenstates closest to $E = 0$ with the analytical predictions. Table I summarizes the results.

Figure 5 illustrates how the spatial profiles of the protected states change with the domain wall position. As the wall moves toward the left edge, $|\mathcal{L}\rangle$ becomes increasingly compressed into a shorter domain (fewer sites available for its exponential tail), while $|\mathcal{R}\rangle$ extends over a longer domain. The state $|\mathcal{S}\rangle$ develops an asymmetric envelope, decaying faster into the shorter domain.

C. Effective couplings and the asymmetry mechanism

The effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ are computed numerically by projecting the full Hamiltonian onto the protected subspace. The results (Fig. 6) confirm the exponential scaling predicted by Eq. (12), with the numerical values listed in Table II.

The coupling asymmetry has a direct and dramatic consequence for the transfer fidelity. In a three-site chain

TABLE I. Protected states for the asymmetric domain wall configurations in an $L = 21$ chain ($v = 0.5, w = 1.0$). $E_{\pm} = \pm\sqrt{J_{\mathcal{LS}}^2 + J_{\mathcal{SR}}^2}$ are the energies of the two split levels of the effective three-site chain [Eq. (9)], the central level remaining pinned at $E = 0$. “Min. overlap” is the smallest singular value of the overlap matrix between the analytical states [Eqs. (6)–(8)] and the numerical eigenstates (unity = perfect agreement). $J_{\mathcal{LS}}/J_{\mathcal{SR}}$ is the effective coupling ratio that sets the Rabi bound of Eq. (17).

j_{DW}	$\ell_1 + \ell_2$	$E_{\pm}$	Min. overlap	$J_{\mathcal{LS}}/J_{\mathcal{SR}}$
11	11 + 9	± 0.024	0.9998	0.50
7	7 + 13	± 0.043	0.9994	8.0
5	5 + 15	± 0.086	0.9978	32.0

with unequal hoppings $J_1 \neq J_2$, the maximum achievable population transfer from site 1 to site 3 is bounded by [3]

$$f_{\text{max}} = \frac{4J_1^2 J_2^2}{(J_1^2 + J_2^2)^2}, \quad (17)$$

which equals unity only when $J_1 = J_2$ and decreases rapidly with the coupling ratio. For our quasi-centered case ($J_{\mathcal{LS}}/J_{\mathcal{SR}} = 0.5$): $f_{\text{max}} = 4 \times 0.25/(1.25)^2 = 16/25 = 0.64$.

D. Transfer fidelity in asymmetric chains

Figure 7 shows the transfer dynamics for the three asymmetric configurations, using the standard $\sin^2$ pulse with $t_{\text{prep}} = 15$. Table III reports the optimal fidelities obtained from a scan over $t_{\text{tr}} \in [30, 600]$.

The quasi-centered case ($j_{\text{DW}} = 11$) achieves $f = 0.637$, in quantitative agreement with the analytical bound $f_{\text{max}} = 0.64$ from Eq. (17). The small discrepancy arises from residual diabatic excitations during the ramp phases. For $j_{\text{DW}} = 7$ and 5, the fidelity drops to $f = 0.061$ and $f = 0.004$, respectively—effectively rendering the transfer useless.

Figure 8 presents a direct comparison of the fidelity-vs- t_{tr} curves for all three wall positions. The oscillation frequencies and amplitudes differ markedly, reflecting the distinct effective coupling landscapes.

E. Systematic study: fidelity vs. wall position

To map out the full dependence of the transfer fidelity on domain wall position, we perform systematic scans for chains of length $L = 11$ and $L = 15$, placing the domain wall at all allowed (odd) sites. Figure 9 shows

TABLE II. Effective couplings for the three domain wall positions ($L = 21, v = 0.5, w = 1.0$).

j_{DW}	$J_{\mathcal{LS}}$	$J_{\mathcal{SR}}$	Ratio	Predicted ratio
11	0.0105	0.0210	0.50	$(v/w)^1 = 0.50$
7	0.0423	0.0053	8.0	$(w/v)^3 = 8.0$
5	0.0862	0.0027	32.0	$(w/v)^5 = 32.0$

TABLE III. Optimal transfer fidelity for the asymmetric domain wall positions ($L = 21$, standard $\sin^2$ pulse, $t_{\text{prep}} = 15$). $t_{\text{tr}}^{\text{opt}}$ is the protocol time of the first fidelity maximum from a scan over $t_{\text{tr}} \in [30, 600]$, and f_{max} the corresponding peak fidelity. The peaks track the analytical Rabi bound of Eq. (17), set by the coupling ratio $J_{\mathcal{LS}}/J_{\mathcal{SR}}$ of Table II.

j_{DW}	$\ell_1 + \ell_2$	$t_{\text{tr}}^{\text{opt}}$	f_{max}
11 (quasi-center)	11 + 9	420	0.637
7 (one-third)	7 + 13	95	0.061
5 (one-quarter)	5 + 15	495	0.004

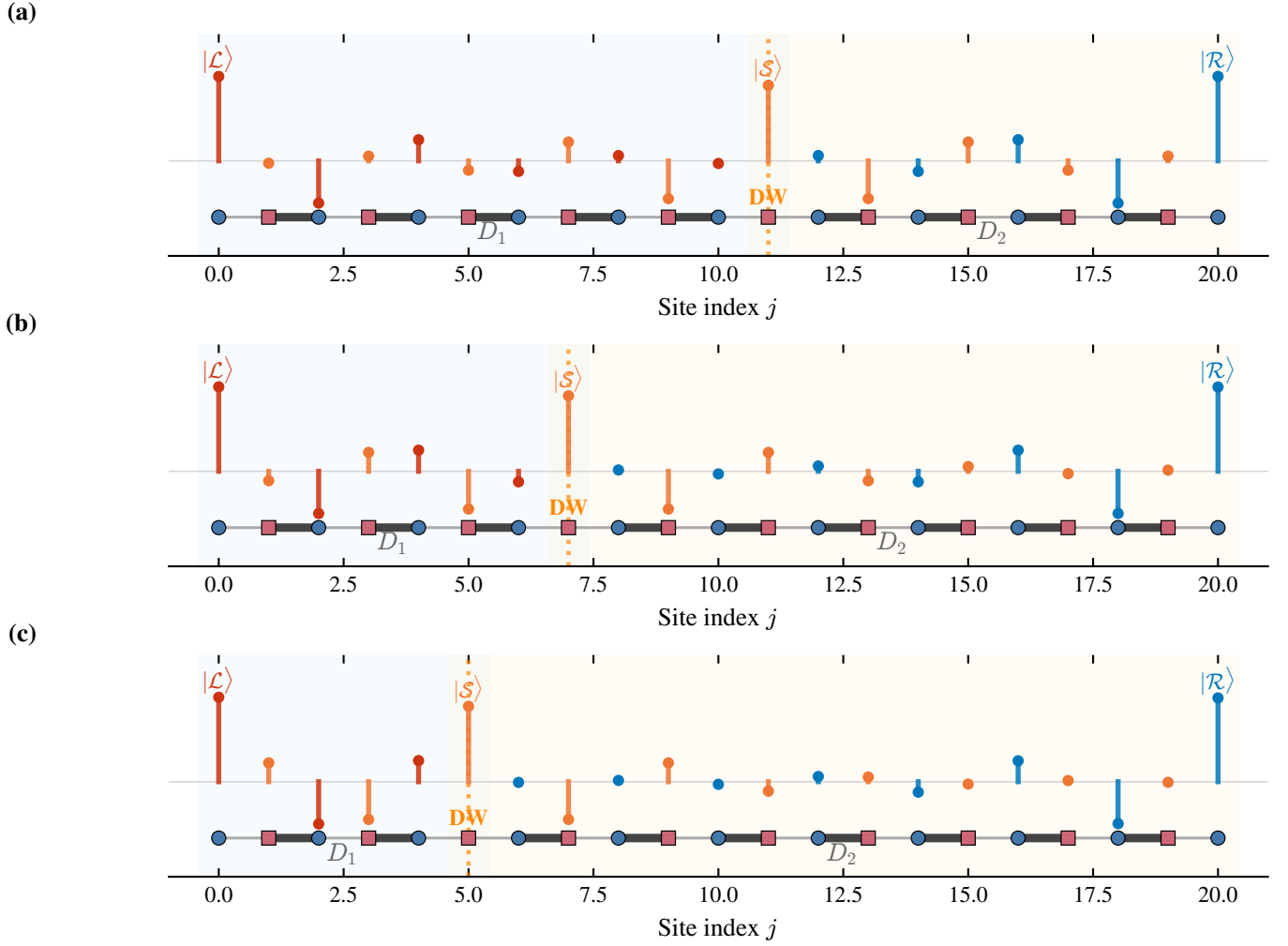


FIG. 5. Spatial profiles of the three protected states $|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, $|\mathcal{R}\rangle$ for different domain wall positions in an $L = 21$ chain: (a) $j_{\text{DW}} = 11$ (quasi-center), (b) $j_{\text{DW}} = 7$ (one-third), (c) $j_{\text{DW}} = 5$ (one-quarter). As the wall moves toward the left edge, $|\mathcal{L}\rangle$ is compressed into the shorter domain while $|\mathcal{R}\rangle$ spreads over the longer one; the exponential localization and sublattice polarization (alternating signs) are preserved in all cases.

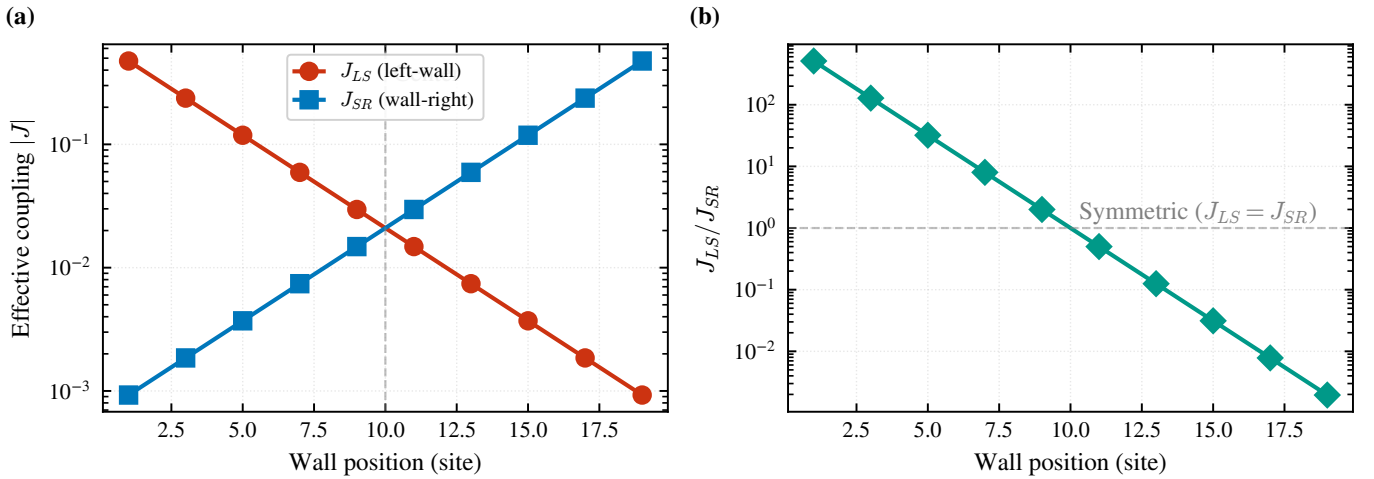


FIG. 6. Effective couplings J_{LS} (left-soliton) and J_{SR} (soliton-right) as a function of the domain wall position for $L = 21$. The coupling asymmetry grows exponentially with the displacement from the chain center, consistent with Eq. (12).

that the fidelity profile is perfectly symmetric about the chain center (reflecting the equivalence of swapping the

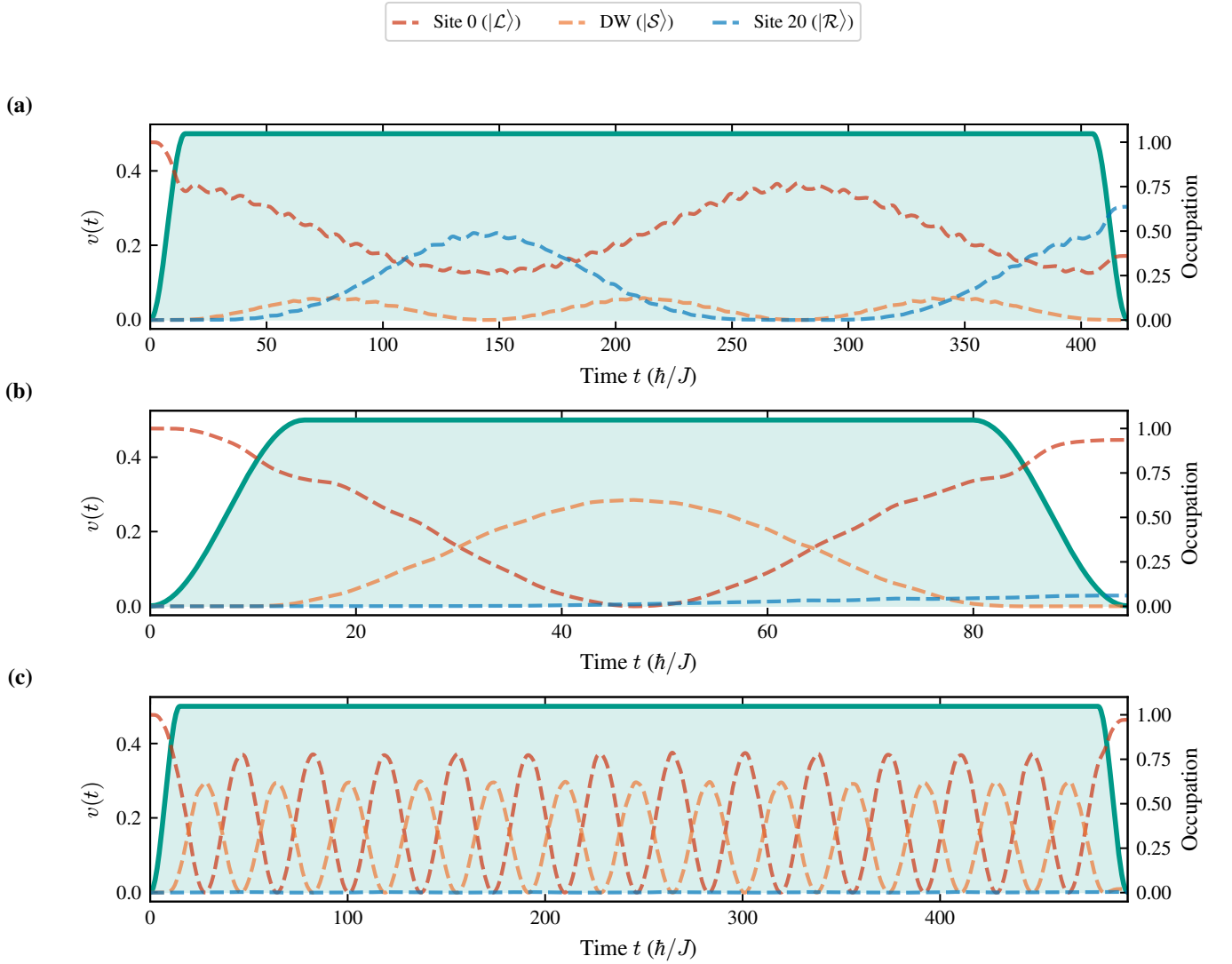


FIG. 7. Transfer dynamics for the three asymmetric domain wall configurations ($L = 21$, standard $\sin^2$ pulse, $t_{\text{prep}} = 15$): (a) $j_{\text{DW}} = 11$, (b) $j_{\text{DW}} = 7$, (c) $j_{\text{DW}} = 5$, each at its optimal t_{tr} (Table III). Each panel shows the shaded ramp $v(t)$ (left axis) and the occupations of the three localized sites—site 0 ($|\mathcal{L}\rangle$), the domain wall ($|\mathcal{S}\rangle$) and site $L - 1$ ($|\mathcal{R}\rangle$) (right axis). As the wall moves off-centre, the coupling asymmetry $J_{\mathcal{LS}} \neq J_{\mathcal{SR}}$ caps the population that can be swapped into $|\mathcal{R}\rangle$: the transfer is essentially complete in (a) but increasingly incomplete in (b) and (c), where the excitation remains trapped in $|\mathcal{L}\rangle$ and $|\mathcal{S}\rangle$.

two domains) and sharply peaked:

- $L = 11$: peak fidelity $f = 0.96$ at $j_{\text{DW}} = 5$ (center), dropping to $f = 0.20$ at $j_{\text{DW}} = 3$ and $f = 0.004$ at $j_{\text{DW}} = 1$.
- $L = 15$: peak fidelity $f = 0.994$ at $j_{\text{DW}} = 7$ (center), dropping to $f = 0.22$ at $j_{\text{DW}} = 5$ and $f = 0.014$ at $j_{\text{DW}} = 3$.

The slightly lower centered value for $L = 11$ ($f = 0.96$) relative to the optimized reproduction of Sec. III C ($f = 0.999$) reflects the coarser $\Delta t_{\text{tr}} = 10$ scan grid used here: the true optimum near $t_{\text{tr}} \approx 45$ falls midway between grid points, so the scan reads f at $t_{\text{tr}} \in \{40, 50\}$ rather than at the exact peak.

The symmetric profiles confirm that the transfer fidelity depends only on the *difference* $|\ell_1 - \ell_2|$ between domain lengths, not on their individual values. The exponential sensitivity to this difference—inherited from Eq. (12)—imposes stringent requirements on the precision of the domain wall placement in experimental implementations.

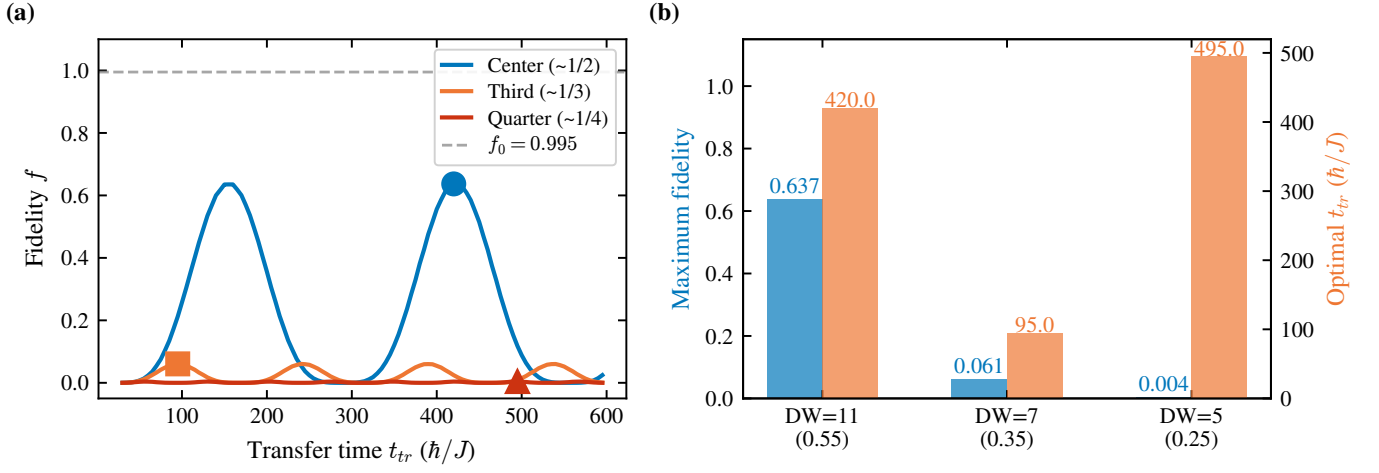


FIG. 8. Transfer fidelity for the three domain wall positions in the $L = 21$ chain. (a) Fidelity f vs. the protocol time t_{tr} for the quasi-centre ($\sim 1/2$), one-third ($\sim 1/3$) and one-quarter ($\sim 1/4$) wall positions; dots mark the first maximum. (b) The corresponding peak fidelity (left axis, bars) and optimal t_{tr} (right axis) for $j_{DW} = 11, 7, 5$. The first fidelity maximum decreases exponentially with the displacement from the centre, consistent with the coupling asymmetry of Eq. (12).

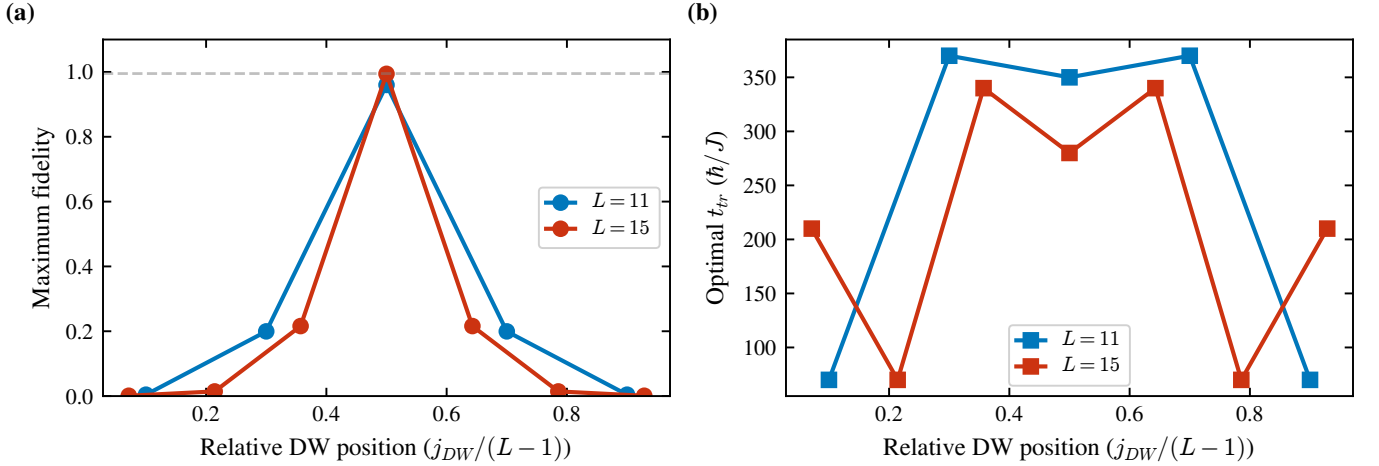


FIG. 9. Maximum transfer fidelity (left panel) and optimal transfer time t_{tr} (right panel) as a function of the normalized domain wall position $j_{DW}/(L-1)$, for chains of length $L = 11$ (blue) and $L = 15$ (orange). Both quantities peak sharply at the chain center ($j_{DW}/(L-1) = 0.5$): the fidelity reaches its maximum and the optimal transfer time its minimum. Only odd-index sites produce valid domain walls.

V. SHORTCUT-TO-ADIABATICITY PROTOCOLS

A. Motivation and theoretical framework

The preparation and readout phases of the transfer protocol involve adiabatic ramps of the intracell hopping $v(t)$. When the ramp time t_{prep} is not sufficiently long compared to the inverse gap $\tau = 2/\Delta$, diabatic excitations to bulk states reduce the fidelity. Shortcut-to-adiabaticity (STA) techniques [10–12, 18] aim to suppress these excitations by modifying the pulse shape, potentially allowing faster operation without sacrificing fidelity.

The exact counterdiabatic Hamiltonian [11]

$$H_{CD}(t) = i\hbar \sum_n (|\dot{n}(t)\rangle\langle n(t)| - \langle n(t)|\dot{n}(t)\rangle |n(t)\rangle\langle n(t)|) \quad (18)$$

cancels all diabatic transitions exactly but requires non-local interactions that are impractical to implement. Instead, we explore *approximate* STA strategies through pulse shape optimization.

B. Pulse profiles

The variants below are *STA-inspired ramp-shaping* strategies—pulse-profile optimizations of the preparation

and readout ramps—rather than exact STA protocols. As such they act only on the ramp phases and, as we show below, cannot overcome the coupling-asymmetry bound, since they do not dynamically control the ratio $J_{\mathcal{LS}}(t)/J_{\mathcal{SR}}(t)$ that governs the transfer phase.

We compare four pulse profiles for the ramp phases (Fig. 10):

1. **Standard $\sin^2$** [Eq. (13)]: The reference protocol from Ref. [1].
2. **Ramp-shaping (α)**: A modified ramp $v(t) = v_{\text{tr}} \sin^{2\alpha}(\Omega t)$ with $\alpha > 1$. The higher exponent produces a steeper ramp that spends less time in the intermediate- v region where the gap is smaller. We test $\alpha = 2$ and $\alpha = 3$.
3. **Global $\sin^2$ ramp**: A single smooth profile over the entire protocol:

$$v(t) = v_{\text{tr}} \sin^2\left(\frac{\pi t}{t_{\text{tr}}}\right), \quad (19)$$

which has no plateau phase and is infinitely differentiable everywhere.

4. **Linear ramp**: $v(t) = v_{\text{tr}} \min(t/t_{\text{prep}}, 1, (t_{\text{tr}} - t)/t_{\text{prep}})$, serving as a baseline with discontinuous $\dot{v}$ that generates stronger diabatic excitations.

C. Results: symmetric chain

Table IV and Fig. 10 summarize the performance of each pulse profile on the symmetric chain ($L = 11$, $\ell = 4$).

The key findings are:

- The standard $\sin^2$ pulse with $t_{\text{prep}} = 15$ provides the optimal balance between speed and fidelity, achieving $f = 0.9994$ at $t_{\text{tr}} = 45$ —reproducing the result of Ref. [1].
- The Global $\sin^2$ ramp achieves the highest absolute fidelity ($f = 0.99995$) but at the cost of a 67% longer protocol time ($t_{\text{tr}} = 75$ vs. 45).

TABLE IV. Pulse comparison for the symmetric chain ($L = 11$, $\ell = 4$, $v_{\text{tr}} = 0.5$, $w = 1.0$).

Pulse	t_{prep}	$t_{\text{tr}}^{\text{opt}}$	f_{max}
Standard $\sin^2$	15	45	0.9994
Standard $\sin^2$	10	89	0.9928
Standard $\sin^2$	5	33	0.9290
Ramp-shaping ($\alpha = 2$)	10	41	0.9706
Ramp-shaping ($\alpha = 3$)	10	43	0.9398
Global $\sin^2$ ramp	—	75	0.99995
Linear	15	97	0.9924

- The Ramp-shaping (α) pulses with reduced $t_{\text{prep}} = 10$ *do not* improve upon the standard protocol. The steeper ramp increases diabatic excitations rather than suppressing them, as the system spends insufficient time adjusting to the changing Hamiltonian.
- The linear ramp performs worst in terms of speed ($t_{\text{tr}} = 97$) due to the discontinuities in $\dot{v}$ at the ramp endpoints, which generate diabatic excitations.

D. Preparation time dependence

Figure 11 shows the dependence of the optimal fidelity on the preparation time t_{prep} for the standard and Ramp-shaping (α) pulses. The fidelity follows a characteristic sigmoid: for $t_{\text{prep}} \ll \tau$ the ramp is too fast and diabatic transitions dominate, while for $t_{\text{prep}} \gtrsim \tau$ the fidelity saturates near its maximum. The crossover occurs at $t_{\text{prep}} \sim 2/\Delta_{\text{min}}$, where Δ_{min} is the minimum spectral gap encountered during the ramp.

E. Results: asymmetric chains with ramp shaping

As a controlled negative test, we ask whether ramp shaping can compensate for the fidelity degradation caused by asymmetric domain-wall placement. We test the standard, Ramp-shaping ($\alpha = 2$), and Global $\sin^2$ ramp on the three asymmetric configurations ($L = 21$, $j_{\text{DW}} = 11, 7, 5$).

The results (Table V and Fig. 12) are unambiguous: *no pulse protocol significantly improves the fidelity in asymmetric configurations*. This confirms that the limitation is not due to diabatic excitations during the ramp phases—which STA protocols are designed to address—but rather stems from the fundamental coupling asymmetry $J_{\mathcal{LS}} \neq J_{\mathcal{SR}}$ in the transfer phase. The pulse shape affects only the preparation and readout fidelity; **the transfer phase itself is governed by the effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$, which are fixed by the domain-wall geometry [Eq. (12)] and not by the ramp profile**. Reshaping the ramp therefore cannot alter the bound f_{max} of Eq. (17): only a protocol that varies the coupling ratio $J_{\mathcal{LS}}(t)/J_{\mathcal{SR}}(t)$ in time—such as

TABLE V. Pulse comparison for asymmetric configurations ($L = 21$). No pulse protocol significantly improves the fidelity. This confirms that the limitation is not due to diabatic ramp excitations, but to the coupling asymmetry in the static Rabi transfer phase.

j_{DW}	Standard	Ramp-shaping ($\alpha = 2$)	Global $\sin^2$
11 (quasi-center)	0.637	0.610	0.609
7 (one-third)	0.061	0.060	0.047
5 (one-quarter)	0.004	0.005	0.003

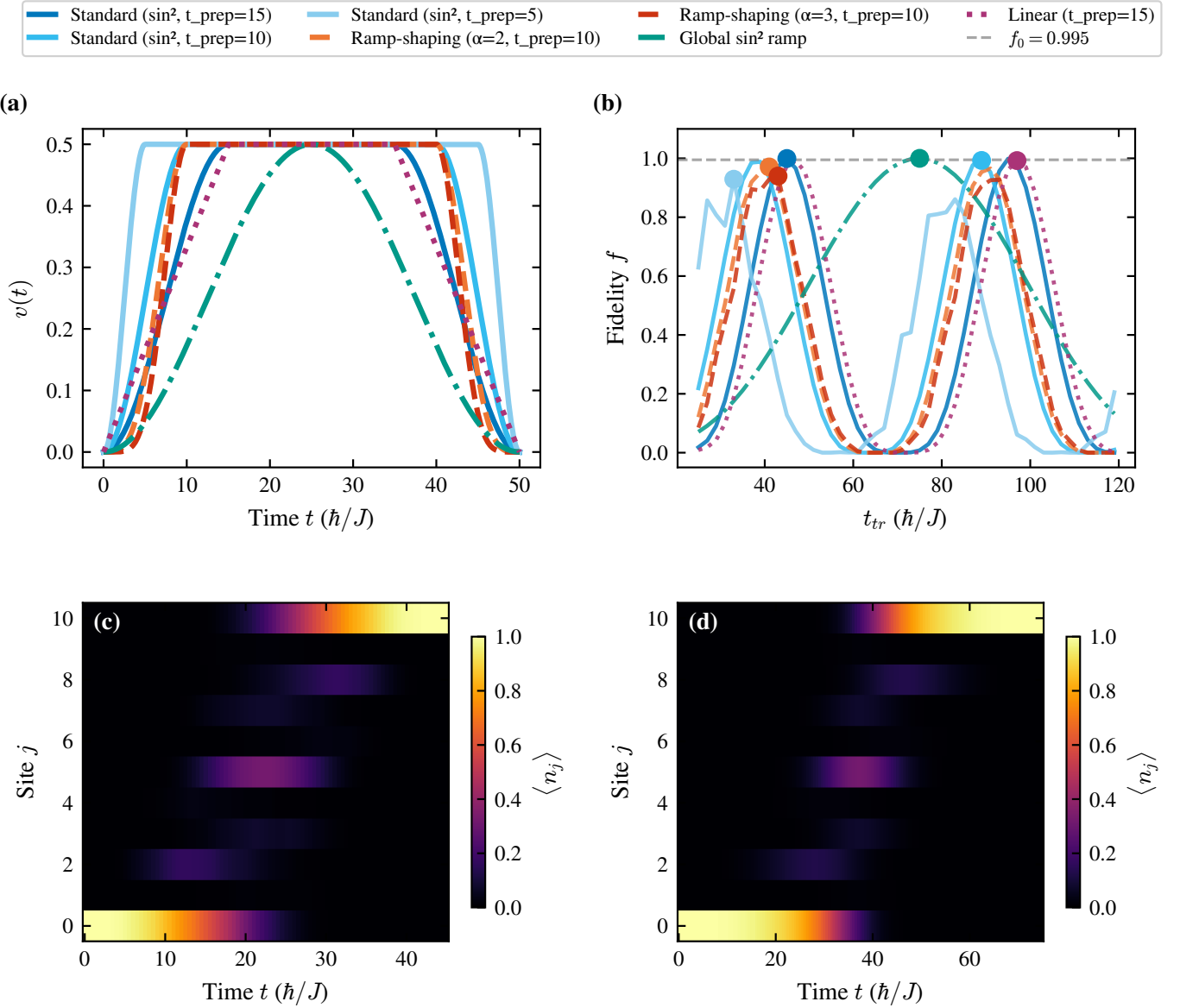


FIG. 10. Comparison of the pulse profiles for the symmetric chain ($L = 11$, $\ell = 4$). (a) The ramp shapes $v(t)$ and (b) the resulting fidelity f vs. t_{tr} for all profiles (shared legend); dots mark each optimum. The standard $\sin^2$ pulse with $t_{\text{prep}} = 15$ (solid blue) gives the best time–fidelity trade-off ($f = 0.9994$ at $t_{tr} = 45$), whereas the Global $\sin^2$ ramp (green dash-dot) reaches the highest absolute fidelity ($f = 0.99995$) at the cost of $t_{tr} = 75$. (c),(d) Site-resolved occupation $\langle n_j \rangle$ vs. time for the standard $\sin^2$ ($t_{tr} = 45$) and the Global $\sin^2$ ramp ($t_{tr} = 75$) protocols, respectively.

the CTAP scheme of Sec. VIII—can overcome the asymmetry.

VI. HYBRID ANALOG-DIGITAL COUNTERDIABATIC PROTOCOL

As an alternative to the domain-wall construction, S. V. Romero *et al.* [13] achieve nonadiabatic edge-state transfer in a *uniform* SSH chain of $2N - 1$ sites. The edge state is transported by sweeping the hopping ratio $t_1(t)/t_2(t)$ through the topological phase diagram—

a Thouless-pump-like protocol [19] whose zero-energy eigenstate migrates from one boundary to the other. Operating adiabatically would require very long protocol times; to go fast, the bare Hamiltonian $H_0(t)$ is supplemented with an approximate counterdiabatic (CD) term derived from the adiabatic gauge potential via the variational method of Sels and Polkovnikov [20].

The essential point for the present work is the structure of this CD correction. To first order it consists of next-nearest-neighbor (NNN) hoppings between sites of the *same* sublattice. Related schemes that accelerate and stabilize SSH edge-state transfer with engineered NNN

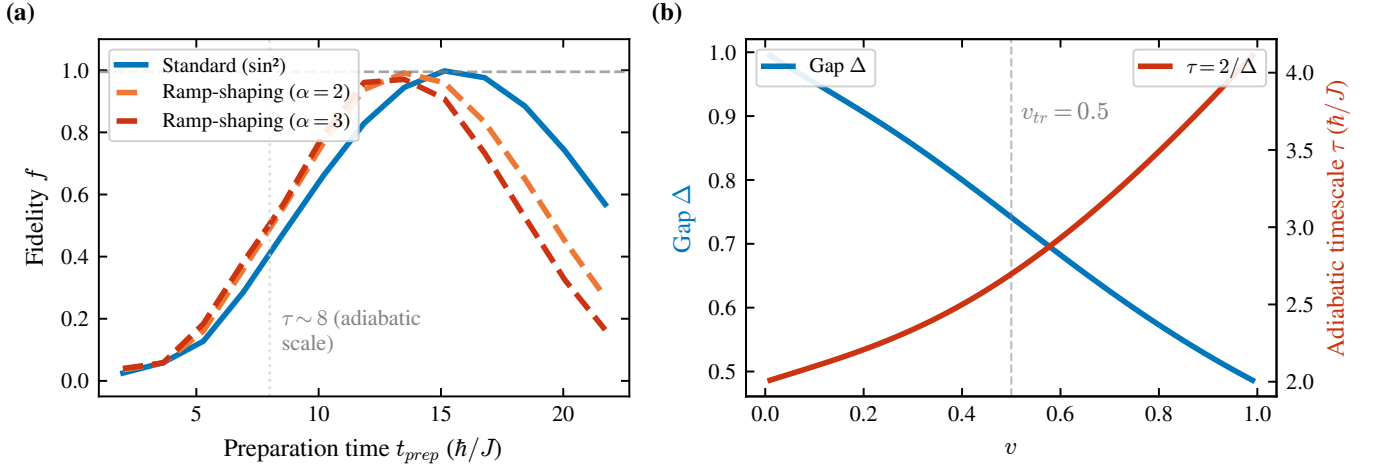


FIG. 11. Optimal transfer fidelity as a function of preparation time t_{prep} for different pulse profiles. The fidelity saturates once t_{prep} exceeds the adiabatic timescale $\tau \sim 2/\Delta_{\text{min}}$.

couplings have also been explored [21]. Since the zero-energy eigenstate has support only on sublattice A , the relevant term reduces to

$$H_{\text{cd}} = -i \sum_{j=1}^{N-1} \kappa_{2j+1,2j-1} c_{2j+1}^\dagger c_{2j-1} + \text{H.c.}, \quad (20)$$

with the couplings κ optimized variationally. *These NNN hoppings break the chiral (sublattice) symmetry of the SSH model*—precisely the symmetry that pins the edge states at $E = 0$ and protects them against off-diagonal disorder. This observation underlies the speed-robustness trade-off analyzed in Sec. VII E.

Higher-order corrections introduce progressively longer-range hoppings and the analytical expansion becomes impractical beyond short chains, so Romero *et al.* optimize the NNN strengths digitally in a hybrid quantum-classical loop. In a fixed nonadiabatic time (a $\sim 100\times$ speed-up over the adiabatic protocol) the optimized driving reaches $F > 0.99$ for chains up to nine sites and $F \sim 0.90$ for fifteen sites under the constraint $\|H_{\text{cd}}\| \sim \|H_0\|$, and they report robustness to on-site and hopping disorder and to Rice-Mele staggering at the $\sim 10\%$ level [13].

In summary, the CD protocol trades the structural simplicity and topological protection of the domain-wall scheme for control complexity: it operates on a defect-free chain at fixed, chain-length-independent speed, but the NNN driving that enables that speed explicitly breaks chiral symmetry—with consequences for disorder robustness that we make precise in Sec. VII E.

VII. COMPARISON OF TRANSFER PARADIGMS

Having analyzed both the domain wall protocol (Secs. III–V) and the hybrid analog-digital CD proto-

col (Sec. VI), we now provide a systematic comparison of both approaches.

A. Physical mechanism

The two protocols exploit fundamentally different physical mechanisms for edge-state transfer:

Domain wall approach (J. Zurita *et al.* [1]). The chain contains topological domain walls that create protected boundary states ($|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, $|\mathcal{R}\rangle$) within the bulk gap. The system is prepared in one boundary state and the hopping amplitude $v(t)$ is ramped adiabatically to populate the target state at the opposite edge. The transfer operates within a *low-energy subspace* spanned by the topologically protected zero-energy states, and the bulk gap provides protection against excitations. The domain wall itself is a static structural feature of the chain.

CD approach (S. V. Romero *et al.* [13]). The chain is a uniform SSH model (no domain wall). The hoppings $t_1(t)$, $t_2(t)$ are swept dynamically, and the transfer occurs by following the instantaneous zero-energy eigenstate of $H_0(t)$ through the topological phase diagram. To accelerate this process without losing fidelity, auxiliary NNN hoppings are added as counterdiabatic corrections that suppress nonadiabatic transitions to other eigenstates. The transfer mechanism is a *dynamical shortcut to adiabaticity* rather than a topological subspace approach.

B. Chain structure

A key practical difference concerns the chain topology:

- The **domain wall protocol** requires careful engineering of the chain structure, placing domain walls at specific positions. As demonstrated in Sec. IV, even small deviations from the ideal wall position

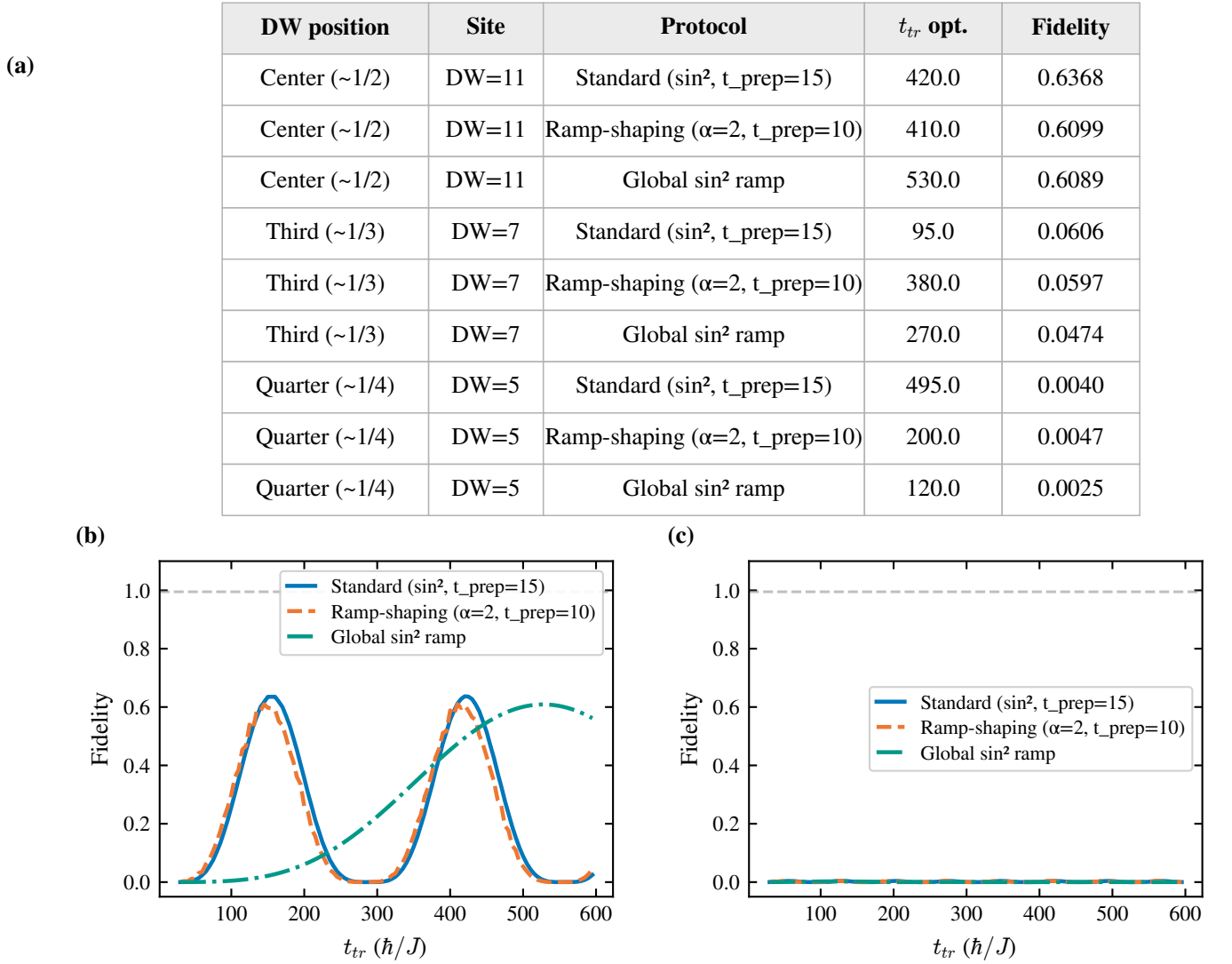


FIG. 12. Combined effect of asymmetry and ramp-shaping pulses ($L = 21$). (a) Optimal protocol time t_{tr}^{opt} and peak fidelity for every combination of wall position ($j_{DW} = 11, 7, 5$) and pulse profile (standard $\sin^2$, Ramp-shaping ($\alpha = 2$), Global $\sin^2$ ramp). (b),(c) Fidelity vs. t_{tr} for the three pulse profiles at the quasi-center ($j_{DW} = 11$, peak $f \approx 0.64$) and at a strongly asymmetric position ($j_{DW} = 5$), respectively. At every position the three pulse variants are essentially indistinguishable and none approaches $f_0 = 0.995$: pulse shaping acts only on the ramps and cannot overcome the coupling asymmetry that governs the transfer phase.

cause exponential degradation of the transfer fidelity [Eq. (17)]. This imposes stringent fabrication requirements.

- The **CD protocol** operates on a uniform SSH chain with no structural defects. The complexity is shifted from the chain architecture to the control fields: time-dependent NNN hoppings $\kappa(t)$ must be engineered between sublattice A sites. These can be tuned continuously during the protocol.

C. Scalability

Both approaches face scalability challenges, but of different nature:

Domain wall protocol. The transfer time scales exponentially with the domain length [Eq. (16)]: $t_{tr} \sim (w/v)^{\ell/2}$. For $w = 2v$ and $\ell = 10$, $t_{tr} \sim 2^5 = 32$ (in units of $1/v$). The multi-domain construction of Ref. [1] mitigates this by dividing the chain into N shorter domains, reducing the transfer time from $(w/v)^{L/2}$ to $N \times (w/v)^{L/(2N)}$ —an exponential improvement. The protocol has been demonstrated numerically for chains up to $L \sim 100$ sites [1], though the number of domain walls must increase proportionally.

CD protocol. The operation time is fixed at $T = \pi/\Omega$ (nonadiabatic regime), which does *not* scale with the chain length. However, the fidelity degrades for longer chains: the approximate CD correction (NNN hoppings only) becomes less accurate as the chain length increases, since higher-order CD corrections—which introduce increasingly long-range hoppings—are needed but are truncated. With the equal-hopping ansatz and the constraint $\|H_{\text{cd}}\| \sim \|H_0\|$, the fidelity drops to $\sim 90\%$ for a 15-site chain [13]. The $H_0 = 0$ variant maintains $F > 99\%$ for 15 sites but requires engineering a purely CD-driven system without the reference Hamiltonian.

Table VI provides a quantitative comparison.

D. Experimental requirements

The domain wall protocol requires only nearest-neighbor hopping control with a single time-dependent parameter $v(t)$. The domain wall structure is *static* and must be encoded at fabrication time. This makes it naturally suited to platforms with fixed lattice geometry, such as quantum dot arrays [22] or photonic lattices [23].

The CD protocol additionally requires time-dependent NNN hopping terms $\kappa(t)$ between sublattice A sites. While more complex to control, this is feasible in several platforms: in superconducting qubit chains, NNN hoppings can be implemented via geometric phases or artificial gauge fields [13]; in cold atoms, angular momentum states [13] or synthetic dimensions provide tunable long-range couplings. The digital optimization via variational circuits provides discretized solutions that are naturally implementable in gate-based quantum processors.

E. Robustness comparison

The two protocols exhibit different robustness characteristics:

- The domain wall protocol enjoys *topological protection*: the boundary states are robust against any perturbation that respects chiral symmetry. However, perturbations that break chiral symmetry (such as on-site disorder or Rice-Mele potentials) can mix the protected subspace with bulk states. Furthermore, as demonstrated in Sec. IV, the protocol is exquisitely sensitive to the *structural* asymmetry of the domain wall position—a form of “geometric disorder” that no pulse optimization can correct.
- The CD protocol is robust against moderate disorder ($\sigma/t_0 \lesssim 0.1$) without relying on topological protection. The robustness arises from the variational optimization, which learns CD driving profiles that are effective even in the presence of perturbations. However, since the protocol deliberately breaks chiral symmetry by introducing NNN

hoppings, it does not benefit from topological protection in the conventional sense.

F. Complementarity of the two approaches

The two paradigms are complementary rather than competitors. The domain wall protocol excels in scenarios where:

- The chain structure can be fabricated with high precision,
- Only nearest-neighbor couplings are available,
- Chiral-symmetry-preserving conditions can be maintained,
- Long chains are needed (via multi-domain construction).

The CD protocol is preferable when:

- The chain structure cannot be precisely controlled,
- NNN couplings are experimentally accessible,
- Fast operation (nonadiabatic regime) is required to beat decoherence,
- A uniform chain is simpler to fabricate than one with domain walls.

A natural future direction is to combine both approaches: using domain wall-mediated transfer as the primary mechanism while adding CD corrections to suppress the residual nonadiabatic excitations during the preparation and readout ramps. This “best of both worlds” strategy would leverage topological protection for the transfer phase while using CD driving for the time-critical ramp phases—precisely the regime where the ramp-shaping variants of Sec. V were shown to provide only marginal improvements.

VIII. CTAP: DYNAMIC CONTROL BEYOND THE STATIC RABI REGIME

The results of Sec. V established that *no ramp-based pulse optimization* can overcome the fidelity degradation caused by coupling asymmetry in asymmetric domain wall configurations. However, this finding must be interpreted carefully. It does not imply that topological protection fails for asymmetric chains: the three edge states ($|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, $|\mathcal{R}\rangle$) remain perfectly well-defined and well-separated from the bulk in all configurations studied. What fails is the specific *transfer protocol*—a static Rabi oscillation scheme that requires equal couplings $J_{\mathcal{LS}} = J_{\mathcal{SR}}$.

In this section we show, following the CTAP (Coherent Tunnelling Adiabatic Passage) framework of

TABLE VI. Comparison of the two transfer protocols. DW = domain wall, CD = counterdiabatic.

Feature	DW protocol [1]	CD protocol [13]
Chain structure	Domain walls at specific sites	Uniform SSH chain
Transfer mechanism	Rabi oscillations in topological subspace	Following instantaneous eigenstate via CD
Interactions	Nearest-neighbor only	NN + NNN (sublattice A)
Number of sites	L odd or even, $N \geq 2$ domains	$2N - 1$ sites (odd)
Transfer time	$t_{\text{tr}} \sim (w/v)^{\ell/2}$, exponential	$T = \pi/\Omega$, fixed (nonadiabatic)
Scalability (time)	Exponential; mitigated by multi-domain	Constant time; fidelity decreases with L
Fidelity ($L = 11$, sym.)	$f = 0.999$	$F > 0.97$ ($H_0 + H_{\text{cd}}$); $F > 0.99$ ($H_0 = 0$)
Sensitivity	Exponential to DW position	No structural sensitivity
Disorder robustness	Topological protection	Robust for $\sigma/t_0 \lesssim 0.1$
Control complexity	Single ramp $v(t)$	Time-dependent NNN $\kappa(t)$ profiles
Optimization	$t_{\text{prep}}, t_{\text{tr}}$	$r(N-1)$ or r parameters
Experimental platforms	QD, SC qubits, cold atoms	Same; requires NNN coupling tunability

A. D. Greentree, J. H. Cole, A. R. Hamilton, and L. C. L. Hollenberg [14], that high-fidelity transfer in asymmetric SSH domain wall systems can be recovered by switching from the static Rabi protocol to a dynamic counter-intuitive pulse sequence. The key insight is that the effective three-level Hamiltonian *always* possesses a zero-energy dark state that mediates direct $|\mathcal{L}\rangle \rightarrow |\mathcal{R}\rangle$ transfer without populating $|\mathcal{S}\rangle$, regardless of the magnitude of $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$.

A. Reframing the asymmetry constraint

Our results demonstrate that the fidelity collapse in asymmetric configurations is *not* a fundamental limitation of topological protection. It is an artifact of the static Rabi protocol: the $\sin^2$ ramp fixes the hopping $v(t)$ uniformly across the entire chain, so both $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ are determined by the same v_{tr} parameter. In asymmetric configurations ($\ell_1 \neq \ell_2$), this leads to an exponentially large coupling ratio $J_{\mathcal{LS}}/J_{\mathcal{SR}} \sim (w/v)^{(\ell_2 - \ell_1)/2}$ [Eq. (12)], for which the Rabi maximum fidelity is bounded by Eq. (17).

The topological protection is unaffected: the edge states are exponentially localized, the gap remains open, and the effective Hamiltonian still has the three-site tight-binding form (9). The asymmetry introduces no instability in the protected subspace—it merely makes the Rabi oscillation amplitude small. The natural solution is therefore to *decouple the control of the left and right domains*, enabling a dynamic protocol in which the coupling ratio $J_{\mathcal{LS}}(t)/J_{\mathcal{SR}}(t)$ is varied in time to exploit the dark state of the effective Hamiltonian.

B. The effective three-level Hamiltonian and its dark state

We now allow the effective couplings to be time-dependent by introducing independent control parameters $v_L(t)$ and $v_R(t)$ for the intracell hopping in the left

and right domains, respectively. The time-dependent effective Hamiltonian in the protected subspace reads

$$\mathcal{H}_{\text{eff}}(t) = \begin{pmatrix} 0 & J_{\mathcal{LS}}(t) & 0 \\ J_{\mathcal{LS}}(t) & 0 & J_{\mathcal{SR}}(t) \\ 0 & J_{\mathcal{SR}}(t) & 0 \end{pmatrix} \quad (21)$$

in the basis $(|\mathcal{L}\rangle, |\mathcal{S}\rangle, |\mathcal{R}\rangle)$, where both couplings now depend on time through the independent domain controls. This is directly analogous to the three-well Hamiltonian of Ref. [14] [their Eq. (1)], with our $J_{\mathcal{LS}}(t)$ and $J_{\mathcal{SR}}(t)$ playing the roles of the tunneling rates $\Omega_{12}(t)$ and $\Omega_{23}(t)$, respectively.

The eigenvalues of Eq. (21) are

$$E_{\pm} = \pm \sqrt{J_{\mathcal{LS}}^2(t) + J_{\mathcal{SR}}^2(t)}, \quad E_0 = 0, \quad (22)$$

so the middle eigenvalue is *always zero*, independent of the magnitudes and ratio of the couplings. The corresponding zero-energy eigenstate (the *dark state*) is [14]

$$|D_0(t)\rangle = \cos \Theta(t) |\mathcal{L}\rangle - \sin \Theta(t) |\mathcal{R}\rangle, \quad (23)$$

where the *mixing angle* $\Theta(t)$ is defined by

$$\tan \Theta(t) = \frac{J_{\mathcal{LS}}(t)}{J_{\mathcal{SR}}(t)}. \quad (24)$$

The crucial property of $|D_0(t)\rangle$ is that it has *zero component on $|\mathcal{S}\rangle$* : the soliton state at the domain wall is completely absent from the dark state. This is a direct consequence of the chiral structure of the effective three-site chain and is preserved at all times regardless of the coupling ratio.

The dark state provides a natural transfer mechanism: if at $t = 0$ we have $J_{\mathcal{LS}} = 0$ and $J_{\mathcal{SR}} \neq 0$, then $\Theta = 0$ and $|D_0\rangle = |\mathcal{L}\rangle$ (the particle is on the left). If at $t = T$ we have $J_{\mathcal{LS}} \neq 0$ and $J_{\mathcal{SR}} = 0$, then $\Theta = \pi/2$ and $|D_0\rangle = -|\mathcal{R}\rangle$ (the particle is on the right). A slow, adiabatic change of $\Theta(t)$ from 0 to $\pi/2$ transfers the population from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$ with perfect fidelity, *without ever populating $|\mathcal{S}\rangle$* .

C. Counter-intuitive pulse sequence

The practical implementation of the dark-state transfer follows the counter-intuitive pulse ordering introduced by A. D. Greentree *et al.* [14] and originally developed in the context of optical STIRAP (Stimulated Raman Adiabatic Passage) [12, 24]. The counter-intuitive aspect is that the *right-domain* coupling $J_{SR}(t)$ must be switched on *before* the left-domain coupling $J_{LS}(t)$ —the opposite of the intuitive ordering.

To see why this is necessary, note that for the system to start in $|D_0\rangle = |\mathcal{L}\rangle$ at $t = 0$, we need $\Theta(0) = 0$, which requires $J_{LS}(0) = 0$. Similarly, $|D_0(T)\rangle = -|\mathcal{R}\rangle$ requires $\Theta(T) = \pi/2$, meaning $J_{SR}(T) = 0$. So J_{SR} must be nonzero first (to define the dark state as $|\mathcal{L}\rangle$) and must go to zero last (when the dark state becomes $-\mathcal{R}\rangle$).

Concretely, we implement the two couplings using Gaussian pulse profiles analogous to Eq. (7) of Ref. [14]:

$$J_{SR}(t) = J_{\max}^R \exp\left[-\frac{(t - t_R)^2}{2\sigma^2}\right], \quad (25)$$

$$J_{LS}(t) = J_{\max}^L \exp\left[-\frac{(t - t_L)^2}{2\sigma^2}\right], \quad (26)$$

where $t_R < t_L$ (the right pulse fires first) and the pulse separation $t_L - t_R$ is a tunable parameter of order σ , chosen per configuration to satisfy the adiabaticity condition (27); the specific values used in each figure are listed in Sec. VIII L. In the SSH model, $J_{SR}(t)$ is controlled by varying $v_R(t)$ in the right domain, and $J_{LS}(t)$ is controlled by $v_L(t)$ in the left domain, both independently.

The critical point is that the peak amplitudes $J_{\max}^R$ and $J_{\max}^L$ *do not need to be equal*. For an asymmetric domain wall with $\ell_1 \neq \ell_2$, the coupling constants scale differently with v_L and v_R [Eq. (10)]. One can simply tune the amplitude of each pulse independently to ensure the desired time-dependence of $\Theta(t)$. The adiabaticity condition $|E_{\pm} - E_0| \gg |\dot{D}_0|$ reads [14]

$$\sqrt{J_{LS}^2 + J_{SR}^2} \gg |\dot{\Theta}| = \frac{|J_{SR}\dot{J}_{LS} - J_{LS}\dot{J}_{SR}|}{J_{LS}^2 + J_{SR}^2}, \quad (27)$$

which can always be satisfied by making the pulses sufficiently long, regardless of the coupling asymmetry.

D. Counterdiabatic driving on the effective three-level system

The CTAP protocol is adiabatic and thus intrinsically slow: the pulse duration must satisfy the adiabaticity condition (27). To accelerate the protocol, one can apply the shortcut-to-adiabaticity strategy directly to the effective three-level Hamiltonian (21).

The exact counterdiabatic (CD) correction for a time-dependent two-level dark-state system can be derived following Berry's method [11, 12]. For the three-level system with a single mixing angle $\Theta(t)$, the result takes a

particularly simple form [12]:

$$\mathcal{H}_{CD}^{\text{eff}}(t) = i\hbar \dot{\Theta}(t)(|\mathcal{L}\rangle\langle\mathcal{R}| - |\mathcal{R}\rangle\langle\mathcal{L}|), \quad (28)$$

where $\dot{\Theta}(t) = (J_{SR}\dot{J}_{LS} - J_{LS}\dot{J}_{SR})/(J_{LS}^2 + J_{SR}^2)$ is the rate of change of the mixing angle. Adding $\mathcal{H}_{CD}^{\text{eff}}$ to $\mathcal{H}_{\text{eff}}(t)$ [Eq. (21)] suppresses all non-adiabatic transitions exactly, allowing perfect transfer in arbitrarily short times.

Physically, $\mathcal{H}_{CD}^{\text{eff}}$ requires a *direct imaginary coupling* between $|\mathcal{L}\rangle$ and $|\mathcal{R}\rangle$. In the microscopic SSH chain, this would correspond to engineering a long-range hopping between site 0 and site $L - 1$ that is purely imaginary—a challenging but not impossible requirement in platforms with gauge-field control (e.g., via Floquet engineering or synthetic dimensions). Alternatively, the CD term can be approximated using local operators via the variational CD approach of Sec. VI, applied now to the effective rather than the full Hamiltonian.

An important advantage of the CTAP+CD approach over the full-chain CD protocol (Sec. VI) is that here the CD term acts on the *three-dimensional* protected subspace rather than the $(2N - 1)$ -dimensional full Hilbert space. This makes the optimization significantly more tractable and the required interactions more local (within the effective subspace).

E. Advantages over the static Rabi protocol in asymmetric systems

The CTAP approach offers three decisive advantages over the static Rabi protocol in the presence of coupling asymmetry:

1. **Asymmetry invariance.** The fidelity of CTAP is independent of the ratio $J_{LS}^{\max}/J_{SR}^{\max}$ as long as the adiabaticity condition is satisfied. Whether $\ell_1 = \ell_2$ or $\ell_1 \ll \ell_2$, the protocol can be tuned to achieve $f \rightarrow 1$ by appropriate choice of pulse duration. This is in stark contrast with the static Rabi protocol, for which $f_{\max} \rightarrow 0$ exponentially as $|\ell_1 - \ell_2| \rightarrow \infty$ [Eq. (17)].
2. **No population of the soliton state.** Because the transfer occurs entirely through the dark state $|D_0(t)\rangle$, the soliton state $|\mathcal{S}\rangle$ is never populated during the transfer. In the original Rabi protocol, $|\mathcal{S}\rangle$ is transiently populated (up to 1/2 population at $t = \pi/(2\sqrt{2}J)$), which can enhance decoherence in platforms where the domain wall site is more exposed to environmental noise.
3. **Topological origin of the dark state.** The existence of the zero-energy dark state $|D_0(t)\rangle$ in Eq. (23) is directly tied to the chiral symmetry of the effective Hamiltonian [Eq. (21)]. The latter is preserved as long as the SSH chain remains in the topological phase, independently of the domain

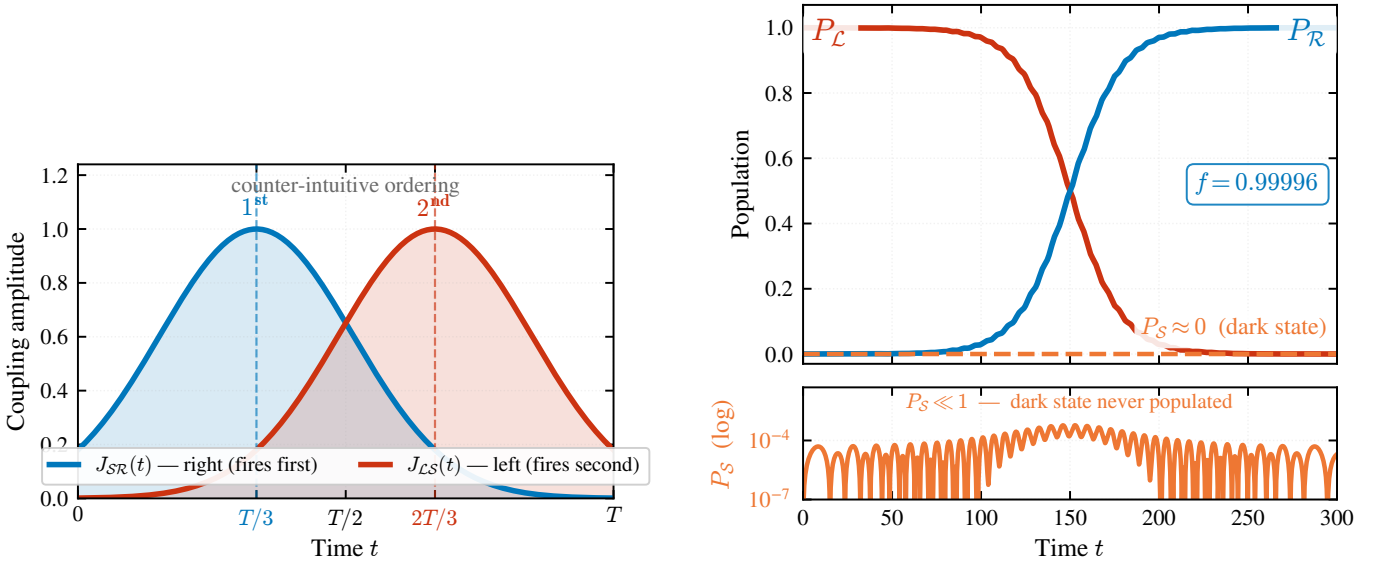


FIG. 13. **(a)** Counter-intuitive CTAP pulse sequence. The right-domain coupling $J_{SR}(t)$ (blue) fires before the left-domain coupling $J_{LS}(t)$ (red), adiabatically rotating the mixing angle $\Theta(t)$ from 0 to $\pi/2$ over the transfer time T . **(b)** Population dynamics of the effective three-level system under the CTAP protocol ($T = 300$, $\sigma = 0.18T$, $J_{\max} = 1$). Starting from $|\mathcal{L}\rangle$ (red), population is transferred to $|\mathcal{R}\rangle$ (blue) with final fidelity $f = 0.9999$. The soliton state $|\mathcal{S}\rangle$ (gold, dashed) remains at zero throughout (lower panel, log scale)—the hallmark of dark-state transfer via the topological zero mode.

wall position. CTAP thus combines the robustness of topological protection with the flexibility of dynamic control.

F. Numerical verification and the physical limit

To substantiate the asymmetry-resilience of CTAP we integrate the effective three-level Hamiltonian (21) with the counter-intuitive Gaussian pulses [Eqs. (25)–(26)] for every valid domain-wall position of the $L = 21$ chain. The peak amplitude of the weaker pulse is set to $1/\rho$, where $\rho = (w/v)^{|\ell_1 - \ell_2|/2}$ is the geometric coupling ratio that also bounds the Rabi protocol [Eq. (17)]; the longer domain physically caps its coupling because the intracell hopping must satisfy $v < w$ to keep the bulk gap open.

Figure 14 reports the result. With well-conditioned pulses, bare CTAP keeps $f \gtrsim 0.99$ across every wall position for which the three-level description remains valid ($\rho \lesssim 10^2$, unshaded region), whereas the static-Rabi fidelity collapses as soon as the wall leaves the center: at $\rho = 32$ ($j_{\text{DW}} = 5$) CTAP exceeds the Rabi bound $f_{\max} = 0.004$ by more than two orders of magnitude. The counterdiabatic term (28) reaches the same fidelity in a fixed, chain-length-independent time; its role is therefore to *accelerate* the transfer, not to raise its fidelity. The residual ceiling of a *fixed* pulse sequence—for example $f \approx 0.91$ at $\rho = 32$ for the pulse separation of Fig. 15—is not a fundamental bound but the boundary-condition effect $f \leq \cos^2 \Theta_0$ analyzed below.

Figure 15 isolates the dependence on the peak ratio ρ

and pins down the origin of the residual infidelity. At fixed pulse separation the converged bare-CTAP fidelity does not remain at unity but follows the analytic ceiling $f = \cos^2 \Theta_0(\rho)$ derived below [Eq. (29)]; the ceiling falls with ρ only because the weaker pulse amplitude $1/\rho$ leaves a larger residual overlap Θ_0 at the protocol endpoints. Re-conditioning the pulses (wider separation, narrower width) drives $\Theta_0 \rightarrow 0$ and restores $f \approx 1$ across the entire range. The genuine physical limit appears only for $\rho \gtrsim 10^2$, where the weaker coupling $J_{\min} \sim 1/\rho$ shrinks the instantaneous gap $\sqrt{J_{LS}^2 + J_{SR}^2}$ until the three protected levels become quasi-degenerate and the effective three-level description—not the CTAP mechanism itself—breaks down. CTAP thus removes the asymmetry bottleneck of the static Rabi protocol across the entire experimentally relevant range; it cannot manufacture coupling where the geometry forbids it.

G. Validation on the full SSH chain

Sections VIII B–VIII F established the CTAP mechanism on the effective three-level Hamiltonian (21). That description, however, retains only the three protected states and discards the remaining $L - 3 = 18$ bulk levels of the physical chain. A legitimate concern is whether the counter-intuitive drive, when applied to the *full* $L = 21$ Hamiltonian (1), still follows the dark state or instead leaks into the bulk continuum that the reduced model cannot see. We therefore integrate the complete time-dependent Schrödinger equation for the three representative wall positions, implementing the two cou-

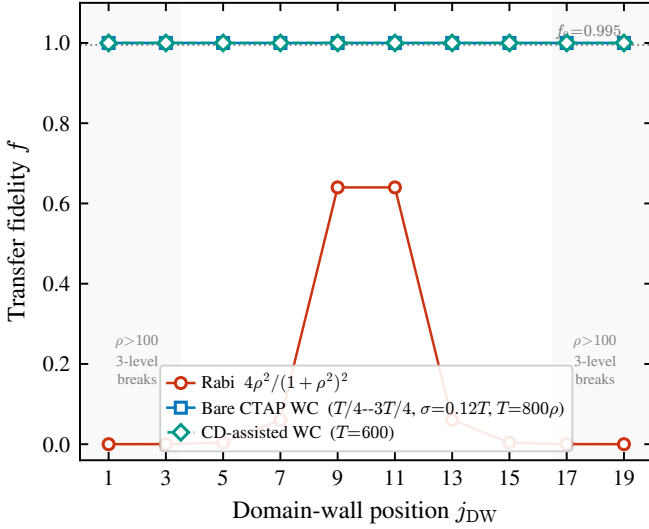


FIG. 14. Transfer fidelity versus domain-wall position j_{DW} for $L = 21$ ($v = 0.5$, $w = 1.0$), from numerical integration of the effective three-level Hamiltonian (21). The static Rabi protocol (red) peaks at $f \approx 0.64$ near the quasi-centre and collapses to $f \lesssim 0.004$ for $j_{\text{DW}} \leq 5$. With well-conditioned pulses [Eqs. (25)–(26); peaks at $T/4$ and $3T/4$, $\sigma = 0.12T$], bare CTAP (blue) and CD-assisted CTAP (teal) both sustain $f \gtrsim 0.99$ at every position where the three-level description holds, and coincide because the counterdiabatic term changes the transfer *time*, not the fidelity. Shaded bands ($j_{\text{DW}} \in \{1, 3, 17, 19\}$, $\rho \gtrsim 10^2$) mark where the weaker coupling closes the protected gap and the three-level model no longer represents the chain. Dotted line: high-fidelity benchmark $f_0 = 0.995$.

plings through the independent domain controls $v_L(t)$ and $v_R(t)$ of Sec. VIII C [pulse parameters $\sigma = 0.15T$, $t_L - t_R = 0.20T$ for $j_{\text{DW}} = 11, 7$ and $\sigma = 0.18T$, $t_L - t_R = 0.36T$ for $j_{\text{DW}} = 5$].

The outcome is summarized in Table VII and Fig. 16. At every wall position the full-chain CTAP fidelity exceeds the high-fidelity benchmark $f_0 = 0.995$, recovering essentially perfect transfer precisely where the static Rabi protocol is useless: $f = 0.9998$ at $j_{\text{DW}} = 11$ (against the Rabi optimum 0.637), $f = 0.9999$ at $j_{\text{DW}} = 7$ (against 0.061), and $f = 0.9995$ at $j_{\text{DW}} = 5$ (against 0.004)—an improvement of up to two and a half orders of magnitude at the strongest asymmetry.

Two features confirm that this is genuine dark-state transport rather than a fortuitous overlap. First, the population outside the three-state protected subspace never exceeds $\sim 10^{-7}$ throughout the protocol: the chiral gap isolates the dark state from the bulk, so the counter-intuitive drive does not excite the continuum. Second, the full-chain fidelity reproduces the effective-model prediction to better than 0.025%, so the 3×3 reduction (21) underlying Figs. 14–15 is not a phenomenological approximation but a quantitatively exact projection of the full dynamics. In particular, the $\cos^2 \Theta_0$ ceiling of Sec. VIII H is a property of the real chain, not of the toy model.

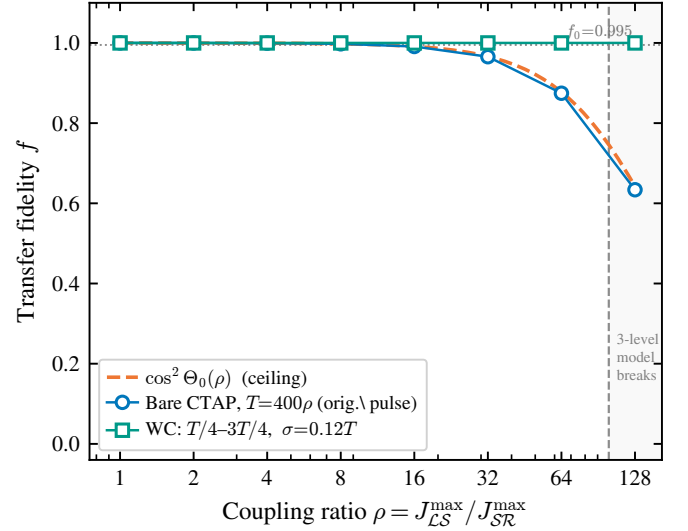


FIG. 15. Transfer fidelity versus the peak-coupling ratio $\rho = J_{\text{max}}^L / J_{\text{SR}}^R$ for the effective three-level system, identifying the residual ceiling as a pulse boundary-condition effect. At the original pulse separation, the converged bare-CTAP fidelity (blue, $T = 400\rho$) tracks the analytic ceiling $\cos^2 \Theta_0(\rho)$ [dashed; Eq. (29)]. Re-conditioned pulses (teal; peaks at $T/4$ and $3T/4$, $\sigma = 0.12T$) drive $\Theta_0 \rightarrow 0$ and keep $f \approx 1$ across the whole range. Beyond $\rho \gtrsim 10^2$ (shaded) the protected gap of the real chain closes and the three-level description ceases to apply. Dotted line: $f_0 = 0.995$.

TABLE VII. Full-chain CTAP on the $L = 21$ SSH Hamiltonian (1) for the three representative wall positions. f_{Rabi} is the static-Rabi optimum (Table III); f_{CTAP} is the converged counter-intuitive-pulse fidelity; the last column is the maximum population leaked outside the three protected states during the protocol.

j_{DW}	(ℓ_1, ℓ_2)	ρ	f_{Rabi}	f_{CTAP}	bulk leakage
11	(11, 9)	2	0.637	0.9998	$\sim 10^{-7}$
7	(7, 13)	8	0.061	0.9999	$\sim 10^{-7}$
5	(5, 15)	32	0.004	0.9995	$\sim 10^{-7}$

The price of asymmetry resilience is paid in time, not fidelity. Adiabaticity through the weaker effective coupling $J_{\text{min}} \sim v(v/w)^{\ell_{\text{max}}/2}$ requires a protocol duration $T \gtrsim J_{\text{min}}^{-1}$ that grows exponentially with the longer (bottleneck) domain: T increases from $\approx 2.6 \times 10^4$ at $j_{\text{DW}} = 11$ to $\approx 6.1 \times 10^4$ at $j_{\text{DW}} = 5$. This exponential cost is intrinsic to a single domain wall and is exactly what the multi-domain construction of Sec. VIII I removes.

H. The fidelity ceiling as a boundary-condition effect

The full-chain results show that the residual infidelity of CTAP at large coupling asymmetry is not a topolog-

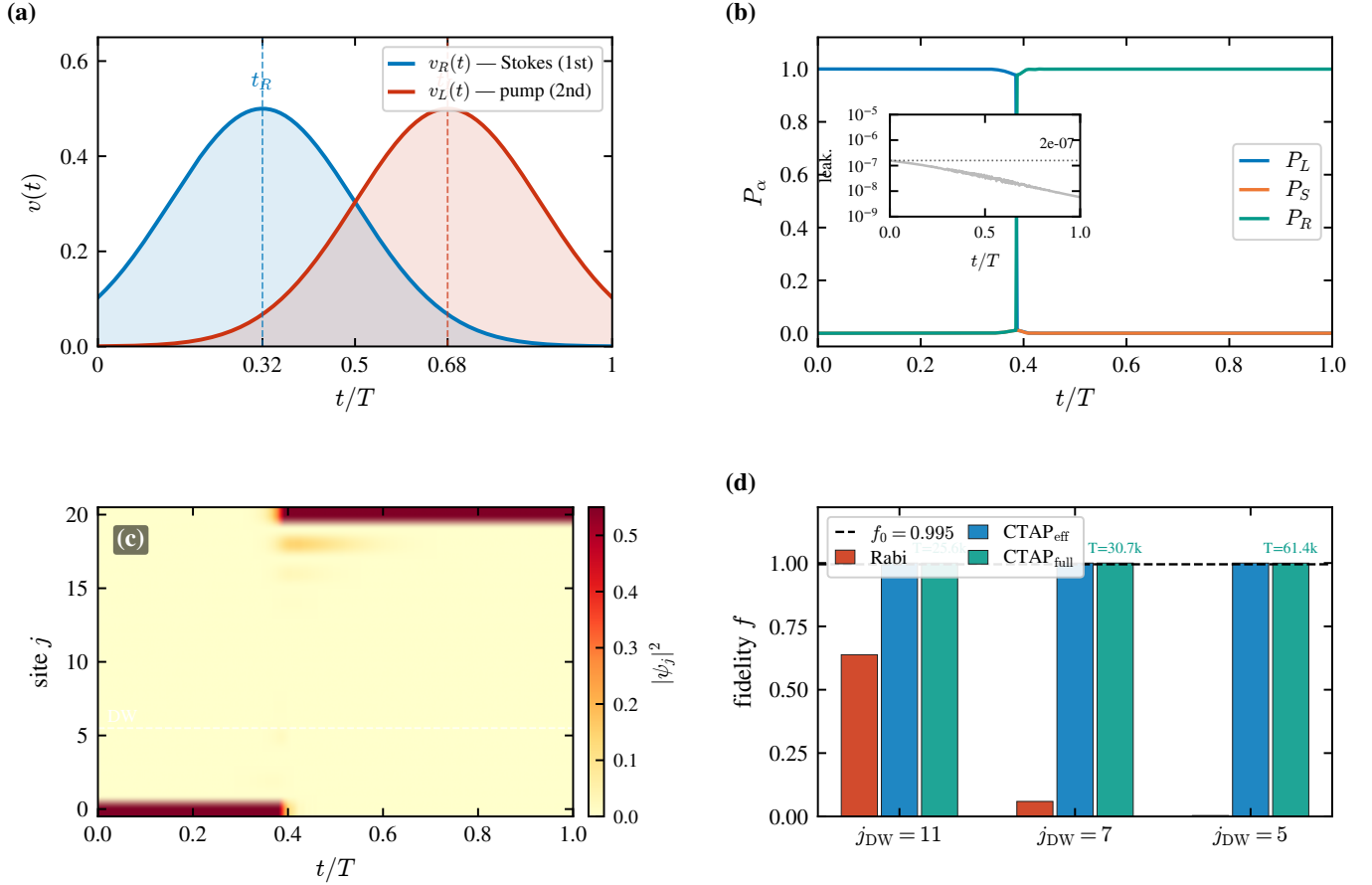


FIG. 16. Validation of CTAP on the full $L = 21$ SSH Hamiltonian (1). **(a)** Counter-intuitive drive in the independent domain controls: the right-domain hopping $v_R(t)$ (Stokes, blue) peaks at $t_R \approx 0.32T$ and fires before the left-domain hopping $v_L(t)$ (pump, red, $t_L \approx 0.68T$), rotating the mixing angle $\Theta(t)$ from 0 to $\pi/2$ (panels (a)–(c) shown for $j_{\text{DW}} = 5$, the strongest asymmetry). **(b)** Population dynamics from the complete time-dependent Schrödinger equation: P_L (red) is transferred to P_R (teal) while the soliton population P_S (orange) stays ≈ 0 . Inset (log scale): the population leaked outside the three protected states never exceeds $\sim 2 \times 10^{-7}$, confirming that the chiral gap isolates the dark state from the bulk continuum. **(c)** Site-resolved density $|\psi_j(t)|^2$: the excitation flows from site 0 to site 20 without accumulating on the domain-wall site (DW, marked)—the spatial signature of dark-state transport. **(d)** Transfer fidelity at the three representative wall positions for the static Rabi optimum (Rabi), the effective three-level CTAP (CTAP_{eff}) and the full-chain CTAP ($\text{CTAP}_{\text{full}}$). The full-chain values reproduce the effective model to better than 0.025% and clear the high-fidelity benchmark $f_0 = 0.995$ (dashed) at every position, whereas the Rabi fidelity collapses away from the centre. Annotated transfer times ($T \approx 2.56 \times 10^4, 3.07 \times 10^4, 6.14 \times 10^4$) grow exponentially with the bottleneck domain.

ical or geometric limitation, but a boundary-condition effect of the pulse sequence. The protocol is initialized in the localized state $|L\rangle$, whereas the instantaneous dark state at $t = 0$ is $|D_0(0)\rangle = \cos \Theta_0 |L\rangle - \sin \Theta_0 |R\rangle$ with $\Theta_0 = \arctan[J_{LS}(0)/J_{SR}(0)]$. The overlap of the initial state with the dark state therefore bounds the achievable fidelity,

$$f \leq \cos^2 \Theta_0, \quad (29)$$

the deficit $\sin^2 \Theta_0$ being the population that fails to follow the dark-state trajectory. This ceiling is set entirely by how completely the pulses suppress $J_{LS}(0)$ (and, symmetrically, $J_{SR}(T)$) at the protocol endpoints; it is removed by increasing the pulse separation, at the cost of a longer protocol time, and is therefore distinct from the

genuine physical limit at $\rho \gtrsim 10^2$, where the protected gap closes and the three-level description breaks down. With suitably conditioned pulses $\Theta_0 \rightarrow 0$ and $f \rightarrow 1$ across the entire experimentally relevant range of asymmetries.

I. Multi-domain acceleration and collective CTAP

The full-chain results of Sec. VIII G show that single-domain-wall CTAP is asymmetry-resilient but pays an exponential price in time: the duration is set by the weakest effective coupling, $T \gtrsim J_{\text{min}}^{-1}$ with $J_{\text{min}} \sim v(v/w)^{\ell_{\text{max}}/2}$, and therefore grows exponentially with the bottleneck domain. The multi-domain construction

of Zurita *et al.* [1] removes this cost. Subdividing a chain of fixed total length into D domains introduces $D - 1$ internal domain walls, each hosting a soliton, so the protected subspace grows to the $D + 1$ states $\{|L\rangle, |S_1\rangle, \dots, |S_{D-1}\rangle, |R\rangle\}$ arranged as a $(D + 1)$ -site effective tight-binding chain. The bottleneck segment shrinks to $\ell_s \simeq L/D$, and since each effective coupling scales as $(v/w)^{\ell_s/2}$, the transfer time collapses exponentially as D increases.

Figure 17(a) quantifies this acceleration for a chain of fixed length ($L \simeq 45$). As D grows from 1 to 4 the bottleneck segment contracts from $\ell_s = 43$ to $\ell_s = 11$, and the estimated transfer time relative to the single-segment reference drops by $T_1/T_D = 916, 4367$, and 13134 for $D = 2, 3, 4$ —almost four orders of magnitude at $D = 4$. The per-segment scaling $J_{\text{bott}} \sim (v/w)^{\ell_s/2}$ is reproduced empirically across the whole range.

The speed-up is only useful if the dark-state mechanism survives the proliferation of intermediate solitons. We verify this by driving the full $D = 4$ chain ($L = 45$, five protected states) with a single counter-intuitive *collective* pulse sequence and integrating the complete time-dependent Schrödinger equation. The collective pulses are shaped directly in the effective couplings $J(t)$ rather than in the bare hopping $v(t)$: a Gaussian profile in v collapses the instantaneous protected gap by up to $\sim 500\times$ at the pulse crossover for the longer segments, whereas shaping in J keeps the gap open and the evolution adiabatic. Figure 17(b) shows the outcome: population flows from $|L\rangle$ to $|R\rangle$ with fidelity $F = 0.9976$, while the inner solitons stay essentially dark— $|S_1\rangle$ and $|S_3\rangle$ remain at ≈ 0 throughout and $|S_2\rangle$ is only transiently populated (peak ≈ 0.24 at $t/T = \frac{1}{2}$), exactly as the odd-site collective dark state predicts. The protocol uses nearest-neighbor couplings only, so it inherits the chiral protection of the single-wall scheme, and it completes $14.3\times$ faster than the $D = 2$ transfer over the same chain. The curves start and end near 0.93 rather than unity because the injected localized site overlaps the protected $|L\rangle$ state at 0.93; this is an initialization overlap, not leakage.

The collective dark state exists because the $D = 4$ configuration has an *odd* number of effective sites ($D + 1 = 5$). Even-site configurations admit no single isolated dark state and require the straddling generalization analyzed in Sec. VIII K.

J. Chiral symmetry and disorder robustness

The defining feature of the SSH edge states is that chiral (sublattice) symmetry pins them at $E = 0$. Because CTAP transports population through these zero modes using nearest-neighbor couplings only, it should inherit that protection; the counterdiabatic protocol of Sec. VI, by contrast, relies on next-nearest-neighbor (NNN) hoppings that break chiral symmetry. We make the contrast quantitative in two steps.

We first ask how the zero mode itself responds to per-

turbations, by diagonalizing the perturbed Hamiltonian and tracking the mean magnitude of its energy $\langle |E_0| \rangle$ [Fig. 18(a)]. Off-diagonal (hopping) disorder—which preserves chiral symmetry—leaves the mode pinned at $E = 0$ to machine precision ($\sim 10^{-16}$) for all disorder strengths. On-site disorder and NNN hoppings, which both break chiral symmetry, instead detach the mode immediately; the NNN term does so linearly, $\langle |E_0| \rangle \approx 0.95\kappa$. The pinning is thus a sharp, symmetry-selective effect: only perturbations that respect chiral symmetry preserve the zero mode.

This spectral pinning carries directly over to the transfer [Fig. 18(b)]. Working at a deep adiabatic margin (so that the disorder-free fidelity exceeds 0.99 in every case) and averaging over disorder realizations (we denote the disorder-averaged transfer fidelity $\langle f \rangle$), we find that off-diagonal disorder barely affects CTAP: $\langle f \rangle = 0.994, 0.994, 0.993$ at $\sigma = 0.05, 0.10, 0.20$. On-site disorder degrades it steadily—0.975, 0.774, 0.302 over the same range—with a large realization-to-realization spread, the signature of the zero mode wandering away from $E = 0$. Adding a moderate NNN coupling $\kappa = J/3$ does *not* erode the off-diagonal robustness: the fidelity stays flat at ≈ 0.99 , because such an NNN term is perturbatively small against the bulk gap (~ 0.5) and shifts the protocol only negligibly.

Together these results sharpen the speed-robustness trade-off between the two paradigms. The protection that survives off-diagonal disorder is exactly the chiral pinning of the zero mode, and CTAP—being nearest-neighbor—retains it in full. The counterdiabatic protocol buys its chain-length-independent speed through NNN driving at strengths $\kappa \sim \|H_0\|$ (Sec. VI), comparable to the hoppings themselves; the spectral panel shows that in that regime the zero mode is strongly detached ($\langle |E_0| \rangle \sim 0.95\kappa$). The acceleration of the CD protocol is therefore obtained at the expense of the chiral pinning that underlies off-diagonal-disorder robustness—a trade-off that CTAP, operating entirely within the nearest-neighbor topological subspace, avoids.

A distinct robustness axis is the placement of the domain wall itself. A ± 2 -site misplacement, which changes the segment lengths and hence the effective couplings, reduces the static-Rabi fidelity to 0.052 but the CTAP fidelity only to 0.419 at fixed, nominally calibrated pulses—roughly an order of magnitude more tolerant. This reflects the asymmetry invariance of Sec. VIII E: the coupling ratio sets only the adiabatic margin of CTAP, not its achievable fidelity, so a positional error that is fatal to the static Rabi protocol merely detunes CTAP. This “geometric” robustness is independent of the symmetry-protection mechanism above.

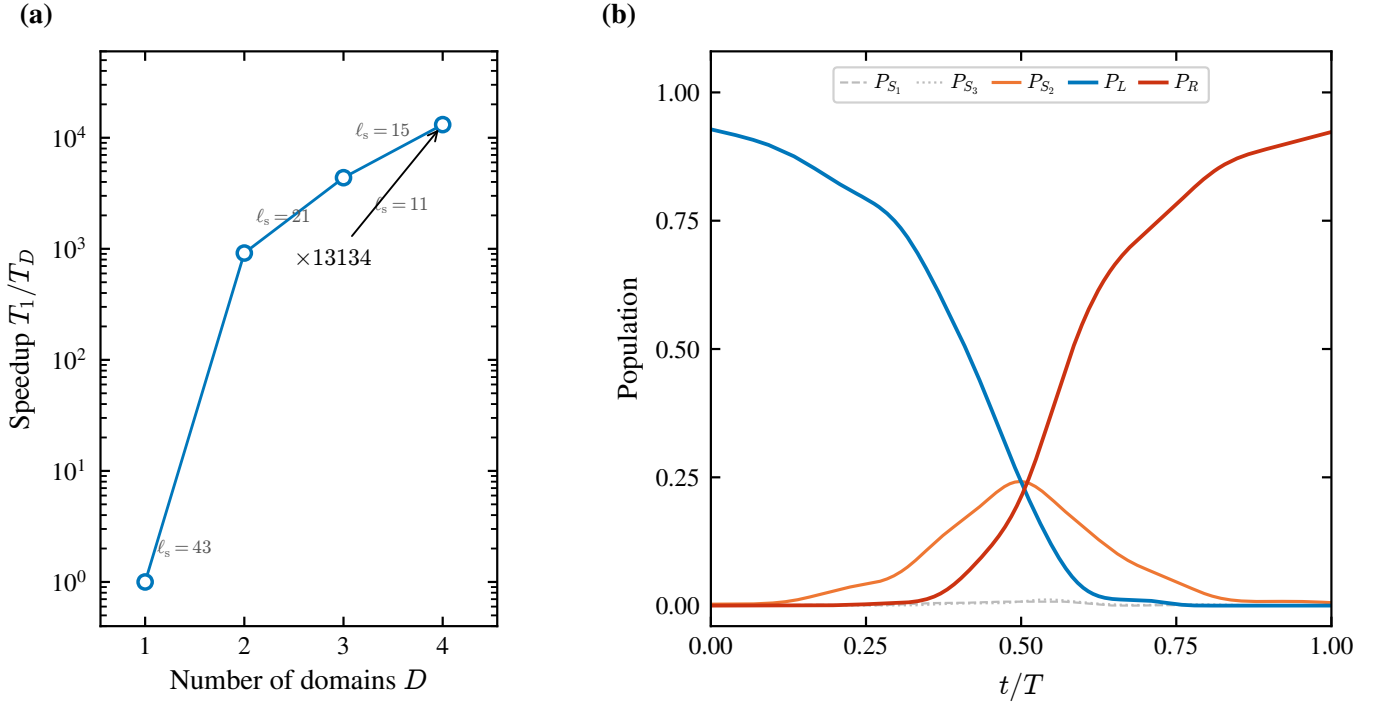


FIG. 17. Multi-domain acceleration of CTAP. (a) Speed-up T_1/T_D of the estimated transfer time as a chain of fixed total length is subdivided into D domains (logarithmic scale). The bottleneck segment length ℓ_s (labels) contracts from 43 at $D = 1$ to 11 at $D = 4$, and the transfer accelerates by up to $\times 13134$. (b) Collective CTAP for the $D = 4$ chain ($L = 45$, five protected states): a single counter-intuitive pulse sequence drives P_L (blue) into P_R (red) with fidelity $F = 0.9976$, while the inner solitons remain dark ($P_{S_1}, P_{S_3} \approx 0$; P_{S_2} , orange, only transiently reaches ≈ 0.24 at $t/T = \frac{1}{2}$). The transfer uses nearest-neighbor couplings only, preserving chiral symmetry.

K. Extension to two domain walls: even-state parity constraint

A natural extension of the domain wall transfer scheme is to consider chains with *two* domain walls, which support *four* topologically protected states: $|\mathcal{L}\rangle$, $|\mathcal{S}_1\rangle$, $|\mathcal{S}_2\rangle$, and $|\mathcal{R}\rangle$. The effective Hamiltonian in this subspace has the structure of a four-site tight-binding chain,

$$\mathcal{H}_{\text{eff}}^{(4)} = \begin{pmatrix} 0 & J_{12} & 0 & 0 \\ J_{12} & 0 & J_{23} & 0 \\ 0 & J_{23} & 0 & J_{34} \\ 0 & 0 & J_{34} & 0 \end{pmatrix}, \quad (30)$$

and one might hope that an analogous CTAP or straddling-CTAP (SCTAP) pulse sequence could transfer population from $|\mathcal{L}\rangle$ to $|\mathcal{R}\rangle$.

However, A. D. Greentree *et al.* [14] identified a fundamental obstacle related to the *parity of the number of effective sites*: the CTAP scheme as described above is only effective for systems with an *odd* number of sites in the effective chain. For an even-site system (such as our four-state chain), there is no isolated zero-energy eigenstate; instead, there are two states whose energies approach zero but never vanish simultaneously (Fig. 6 of Ref. [14]).

Physically, the absence of an isolated dark state means

that any adiabatic transfer trajectory must pass through a region of near-degeneracy between the two low-energy states. Satisfying the adiabaticity condition $|E_0 - E_{\pm}| \gg |\langle \dot{D}_0 | D_{\pm} \rangle|$ in this region requires an *exponentially slow* protocol, negating the advantage of the CTAP approach.

To circumvent this parity constraint, A. D. Greentree *et al.* [14] introduced the *Straddling CTAP* (SCTAP) scheme, which employs an augmented Hamiltonian with three different pulse amplitudes:

$$\mathcal{H}_{\text{SCTAP}} = \begin{pmatrix} 0 & \Omega_1 & 0 & 0 \\ \Omega_1 & 0 & \Omega_S & 0 \\ 0 & \Omega_S & 0 & \Omega_2 \\ 0 & 0 & \Omega_2 & 0 \end{pmatrix}. \quad (31)$$

Here Ω_1 and Ω_2 are the counter-intuitive “outer” pulses (analogous to the CTAP $J_{\mathcal{L}\mathcal{S}}$ and $J_{\mathcal{S}\mathcal{R}}$), and Ω_S is a persistent “straddling” pulse that remains on throughout the protocol. The zero-energy eigenstate of this Hamiltonian [Greentree Eq. (11)] has a modified form that still avoids populating the intermediate sites, but only in the limit $X = \Omega_1\Omega_2/(\Omega_S\sqrt{\Omega_1^2 + \Omega_2^2}) \ll 1$.

For a two-domain-wall SSH chain, the SCTAP approach would translate into:

1. A persistent nonzero hopping v_S coupling the two inner soliton states $|\mathcal{S}_1\rangle$ and $|\mathcal{S}_2\rangle$, which can be im-

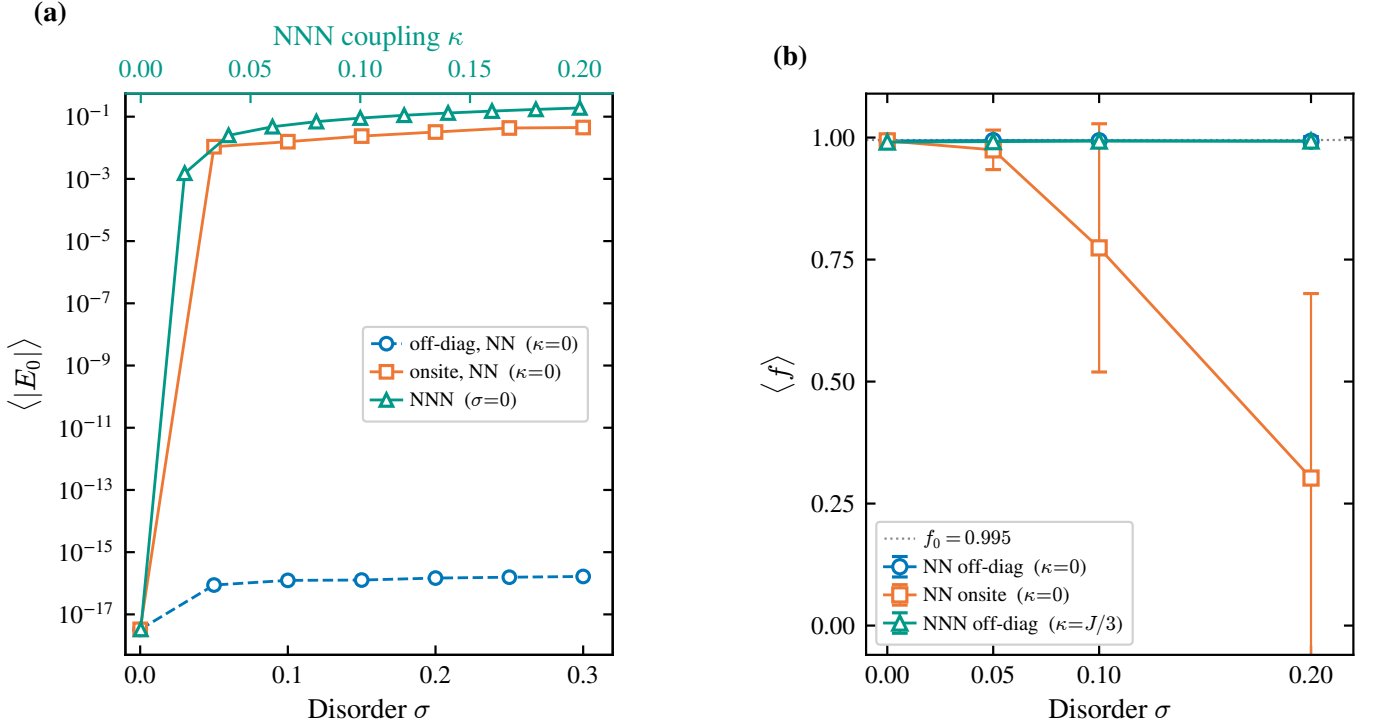


FIG. 18. Chiral symmetry and disorder robustness. (a) Mean zero-mode energy $\langle |E_0| \rangle$ (log scale) under off-diagonal (hopping) disorder σ at $\kappa = 0$ (blue, bottom axis), on-site disorder σ at $\kappa = 0$ (orange, bottom axis), and an NNN coupling κ at $\sigma = 0$ (teal, top axis). Off-diagonal disorder respects chiral symmetry and pins the mode at $E = 0$ to machine precision ($\sim 10^{-16}$); on-site disorder and NNN coupling break it and detach the mode, the latter linearly as $\langle |E_0| \rangle \approx 0.95\kappa$. (b) Mean CTAP transfer fidelity $\langle f \rangle$ versus disorder strength σ (error bars: standard deviation over realizations). Off-diagonal disorder leaves the fidelity flat at ≈ 0.99 (blue), and it stays flat when a moderate NNN coupling $\kappa = J/3$ is added (teal); on-site disorder degrades it to ≈ 0.30 at $\sigma = 0.20$ (orange). Dotted line: $f_0 = 0.995$.

plemented by leaving the central domain in a state of partial dimerization.

2. A counter-intuitive pair of outer pulses $v_R(t)$ (right domain, fires first) and $v_L(t)$ (left domain, fires second), analogous to the one-domain-wall CTAP.

A full analysis of this scheme, including the optimization of Ω_S and verification of the $X \ll 1$ condition for different chain geometries, is deferred to future work.

L. Numerical implementation

Effective couplings by projection. $J_{\mathcal{L}\mathcal{S}}(t)$ and $J_{\mathcal{S}\mathcal{R}}(t)$ [Eqs. (25)–(26)] are computed at each time step by `effective_couplings()` in `ctap_full_chain.py`: the three eigenstates of $H(t)$ with the smallest $|E|$ are identified via `np.argsort(|E|)[:3]` (no explicit energy cutoff; the threshold $|E| < 0.1$ appears only in the unit-test counter, not in the production routines), labeled $|\mathcal{L}\rangle, |\mathcal{S}\rangle, |\mathcal{R}\rangle$ by the position of their probability-density maximum, rotated to the maximally localized (Wannier) basis by diagonalizing the position operator restricted to the three-dimensional subspace, and $J_{\mathcal{L}\mathcal{S}}, J_{\mathcal{S}\mathcal{R}}$ are read

off as the off-diagonal elements of the full Hamiltonian projected onto that localized basis.

Bulk leakage. At every time step `metrics()` projects $|\psi(t)\rangle$ onto the instantaneous protected triplet and computes $1 - (P_{\mathcal{L}} + P_{\mathcal{S}} + P_{\mathcal{R}})$; figures report the *maximum* of this quantity over the full trajectory (`lk.max()` in `fig4_fullchain_ctap.py`, line 86).

Disorder averaging. The spectral-pinning panel [Fig. 18(a)] averages over $N_{\text{real}} = 200$ disorder realizations (`paso4_chiral_spectral.py`); the transfer-fidelity panel [Fig. 18(b)] uses $N_{\text{real}} = 40$ realizations (`paso4_fig_data.py`).

CTAP pulse parameters and time integration. The Gaussian pulse parameters [Eqs. (25)–(26)] differ by figure:

- *Figs. 13 (demo, effective three-level, $T = 300$):* $\sigma = 0.18T$, $t_R = T/3$, $t_L = 2T/3$, $t_L - t_R = T/3 \approx 1.85\sigma$ (`ctap_figures.py`).
- *Fig. 14 (f vs. j_{DW} , effective three-level, `fig15-wc-ctap.py`):* well-conditioned (WC) pulse, $\sigma = 0.12T$, $t_R = T/4$, $t_L = 3T/4$ (separation $T/2$); bare CTAP $T = 800\rho$ (scales with coupling ratio per position), CD-assisted $T = 600$ fixed.

- *Fig. 15* (f vs. ρ , `fig16_ctap_asymmetry_new.py`): two pulse variants. Original pulse ($\sigma = 0.18T$, $t_R = T/3$, $t_L = 2T/3$, $T = 400\rho$) as comparison curve tracking $\cos^2 \Theta_0(\rho)$. WC pulse ($\sigma = 0.12T$, $t_R = T/4$, $t_L = 3T/4$, $T = 50000$ fixed) confirming $f \approx 1$ across all ρ .
- *Table VII / Fig. 16* (full-chain, $L = 21$): form (a) for $j_{\text{DW}} = 11, 7$: $\sigma = 0.15T$, $t_R = 0.40T$, $t_L = 0.60T$ ($t_L - t_R = 0.20T$); form (b) for $j_{\text{DW}} = 5$: $\sigma = 0.18T$, $t_R = 0.32T$, $t_L = 0.68T$ ($t_L - t_R = 0.36T$). $T = \tau_{\text{ad}}/J_{\text{bott}}$ with $\tau_{\text{ad}} \in \{180, 300, 450\}$ (minimum τ_{ad} satisfying $f > 0.995$ and $P_S^{\text{max}} < 0.02$) (`paso3d_final.py`).

All full-chain dynamics are integrated with `scipy.integrate.solve_ivp` (RK45, $\text{rtol} = 10^{-6}$, $\text{atol} = 10^{-8}$); effective three-level dynamics use DOP853 ($\text{rtol} = 10^{-11}$, $\text{atol} = 10^{-13}$, $\Delta t_{\text{max}} = T/4000$).

IX. DISCUSSION AND CONCLUSIONS

We have presented a detailed study of quantum state transfer in SSH chains, comparing two distinct paradigms: domain wall-mediated transfer following J. Zurita *et al.* [1] and hybrid analog-digital counterdiabatic driving following S. V. Romero *et al.* [13], and proposing CTAP as the path beyond the limitations of the static Rabi regime. For the symmetric two-domain configuration ($L = 11$, $N = 2$, $\ell = 4$), we faithfully reproduce the results of Ref. [1]: the optimal transfer time $t_{\text{tr}} = 45$ agrees with the analytical prediction $t_{\text{tr}} \approx 45.6$ and the fidelity $f = 0.999$ exceeds the high-fidelity benchmark $f_0 = 0.995$, providing a validated numerical platform for the subsequent analysis. Our five main conclusions are:

1. Static Rabi transfer fails under coupling asymmetry. Displacing the domain wall from the chain center generates an exponentially large coupling ratio $J_1/J_2 \sim (w/v)^{|\ell_1 - \ell_2|/2}$, which bounds the static Rabi fidelity by $f_{\text{max}} = 4J_1^2 J_2^2 / (J_1^2 + J_2^2)^2$. A single unit-cell displacement in $L = 21$ reduces the fidelity from ~ 1 to 0.637; larger displacements make the transfer negligible. This is *not a topological constraint*: the three edge states ($|\mathcal{L}\rangle$, $|\mathcal{S}\rangle$, $|\mathcal{R}\rangle$) remain exponentially localized and well-separated from the bulk at all configurations studied. The limitation is an artifact of the static Rabi protocol, which couples both domains through a single hopping parameter $v(t)$ and thus cannot independently tune $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$.

2. Ramp shaping cannot overcome the coupling-asymmetry bound. STA-inspired ramp shaping (Global $\sin^2$ ramp, modified-exponent Ramp-shaping (α), and linear ramps) provides marginal improvements in the symmetric case—the Global $\sin^2$ ramp achieves $f = 0.99995$ —but is completely ineffective against coupling asymmetry: no ramp variant reduces the infidelity

by more than statistical noise at any asymmetric wall position. Ramp shaping acts only on the preparation and readout phases; the transfer phase is governed by the fixed effective couplings $J_{\mathcal{LS}}$ and $J_{\mathcal{SR}}$ set by the domain geometry [Eq. (12)], which no envelope choice can alter.

3. CTAP exploits the topological dark state to bypass the asymmetry. Introducing independent controls $v_L(t)$ and $v_R(t)$ for the two domains replaces the static Rabi oscillation with adiabatic evolution along the zero-energy dark state $|D_0(t)\rangle = \cos \Theta(t) |\mathcal{L}\rangle - \sin \Theta(t) |\mathcal{R}\rangle$ —a consequence of chiral symmetry that holds regardless of the coupling ratio. Full-chain integration of the $L = 21$ SSH chain with the counter-intuitive Gaussian pulse sequence recovers $f > 0.999$ at every wall position where the Rabi fidelity collapses to $\lesssim 10^{-2}$, with bulk leakage below 10^{-7} and an effective three-level reduction accurate to better than 0.025%; the residual infidelity is a removable pulse boundary-condition effect ($f \leq \cos^2 \Theta_0$), not a fundamental limit, and the only genuine bound ($\rho \gtrsim 10^2$) arises from geometric gap closing. The effective counterdiabatic correction $\mathcal{H}_{\text{CD}}^{\text{eff}} = i\hbar \dot{\Theta}(|\mathcal{L}\rangle \langle \mathcal{R}| - |\mathcal{R}\rangle \langle \mathcal{L}|)$, acting on the three-dimensional protected subspace, enables perfect transfer in arbitrarily short times and is far more tractable than full-chain CD optimization.

4. Multi-domain subdivision gives exponential acceleration with a collective dark state. Subdividing the chain into D domains at fixed total length shrinks the bottleneck segment and reduces the transfer time by more than four orders of magnitude ($\sim 1.3 \times 10^4$ at $D = 4$). A single counter-intuitive collective pulse sequence achieves $F = 0.998$ across the four-domain chain, keeping the inner soliton states dark and using nearest-neighbor couplings only—so the speed-up is obtained without sacrificing the chiral symmetry that underlies topological protection.

5. Chiral symmetry, not topology alone, governs disorder robustness. Chiral symmetry pins the zero-energy edge states exactly at $E = 0$: off-diagonal (hopping) disorder leaves the zero mode at machine precision and the CTAP fidelity flat at $\langle f \rangle \approx 0.99$, whereas on-site disorder and next-nearest-neighbor hoppings break the symmetry and linearly detach the mode ($\langle |E_0| \rangle \approx 0.95\kappa$ for NNN coupling κ), degrading the transfer. Because CTAP uses only nearest-neighbor couplings it fully inherits this chiral protection, while the NNN-based CD protocol of Ref. [13] sacrifices it to gain speed: the control protocol, not the topological phase alone, determines whether transfer is robust. CTAP is also $\sim 8\times$ more tolerant to domain-wall misplacement than the static Rabi protocol.

The CD protocol of Ref. [13]—achieving fidelities above 99% for chains up to nine sites and $\sim 90\%$ for fifteen sites at a $100\times$ speed-up over the adiabatic protocol—constitutes a complementary paradigm suited to uniform chains with tuneable NNN interactions and stringent time constraints, at the cost of the chiral protection highlighted in conclusion 5. For chains with

two domain walls, the even-state parity constraint precludes a straightforward single-dark-state CTAP protocol; the Straddling-CTAP scheme of Ref. [14], combining a persistent inter-soliton coupling with counter-intuitive outer pulses, offers a route forward whose optimization and experimental verification we leave for future work. Together, these results demonstrate that topolog-

ical quantum state transfer can be made simultaneously fast, asymmetry-resilient, and disorder-robust by choosing control protocols that respect the chiral symmetry of the underlying lattice—a design principle applicable across superconducting qubit, cold-atom, and quantum dot platforms.

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